WYNE

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LOAD THE NECESSARY PACKAGES

```
library("dplyr")
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

SET THE WORKING DIRECTORY

```
setwd("C:/Users/Admin/Desktop/DML")
```

LOAD THE WORKING DIRECTORY

```
df<-read.csv("dplyr_conditional_project_dataset.csv")
View(df)
```

```
colnames(df)
```

```
## [1] "Student_ID"            "Gender"                "Department"
## [4] "Age"                   "Assignment_Score"      "Exam_Score"
## [7] "Attendance_Percentage"
```

```
library(dplyr)

# Example: Total Score = 40% Assignment + 60% Exam
df <- df %>%
  mutate(Total_Score = 0.4 * Assignment_Score + 0.6 * Exam_Score)
```

CREATE THE GRADE COLLUMN

```r
df<-df%>%
  mutate(Grade=case_when(
    Total_Score>=80~"A",
    Total_Score>=70~"B",
    Total_Score>=60~"C",
    TRUE           ~"F"#this handles everything below 60
  ))

print(df)
```

```
##     Student_ID Gender   Department Age Assignment_Score Exam_Score
## 1            1   Male Data Science  27               80         57
## 2            2 Female           IT  24               74         61
## 3            3   Male     Business  29               58         36
## 4            4   Male  Engineering  26               87         51
## 5            5   Male           IT  24               55         67
## 6            6 Female Data Science  18               42         43
## 7            7   Male     Business  18               59         77
## 8            8   Male           IT  26               63         82
## 9            9   Male Data Science  28               93         73
## 10          10 Female Data Science  26               95         76
## 11          11   Male Data Science  21               72         60
## 12          12   Male           IT  26               63         84
## 13          13   Male           IT  20               91         59
## 14          14   Male           IT  24               50         58
## 15          15 Female  Engineering  23               88         47
## 16          16   Male           IT  25               47         94
## 17          17 Female           IT  28               75         41
## 18          18 Female           IT  26               77         91
## 19          19 Female  Engineering  22               79         70
## 20          20   Male           IT  18               59         79
## 21          21 Female Data Science  20               74         54
## 22          22   Male           IT  27               87         99
## 23          23 Female Data Science  29               64         42
## 24          24 Female Data Science  25               74         50
## 25          25 Female Data Science  28               64         48
## 26          26 Female  Engineering  23               68         49
## 27          27 Female  Engineering  25               57         66
## 28          28 Female  Engineering  26               85         97
## 29          29 Female           IT  21               57         85
## 30          30 Female Data Science  18               41         59
## 31          31   Male  Engineering  18               93         92
## 32          32   Male  Engineering  27               74         97
## 33          33 Female           IT  21               55         96
## 34          34 Female  Engineering  29               80         56
## 35          35 Female           IT  24               75         92
## 36          36   Male     Business  19               72         92
## 37          37 Female  Engineering  20               43         83
## 38          38   Male     Business  18               72         86
## 39          39   Male           IT  22               53         76
```

```
## 40          40   Male Data Science  18              60         49
## 41          41   Male  Engineering  25              87         88
## 42          42   Male           IT  18              59         94
## 43          43 Female Data Science  28              95         42
## 44          44 Female Data Science  18              47         87
## 45          45 Female     Business  19              46         94
## 46          46 Female           IT  19              42         39
## 47          47 Female Data Science  29              56         40
## 48          48   Male  Engineering  23              72         81
## 49          49 Female  Engineering  24              87         89
## 50          50 Female     Business  22              51         74
## 51          51   Male Data Science  18              98         86
## 52          52 Female           IT  18              90         50
## 53          53   Male  Engineering  20              61         47
## 54          54 Female Data Science  19              94         64
## 55          55   Male Data Science  22              61         53
## 56          56 Female     Business  29              85         51
## 57          57 Female  Engineering  27              69         97
## 58          58   Male  Engineering  23              77         53
## 59          59   Male           IT  24              77         92
## 60          60   Male  Engineering  21              84         89
## 61          61   Male           IT  24              90         96
## 62          62   Male  Engineering  28              93         57
## 63          63   Male     Business  25              47         43
## 64          64   Male  Engineering  28              66         46
## 65          65   Male           IT  18              66         35
## 66          66 Female           IT  23              73         92
## 67          67 Female     Business  25              60         35
## 68          68   Male  Engineering  22              69         68
## 69          69 Female  Engineering  21              72         82
## 70          70 Female     Business  19              67         35
## 71          71 Female  Engineering  23              86         50
## 72          72 Female  Engineering  23              72         95
## 73          73   Male           IT  28              44         98
## 74          74 Female  Engineering  18              99         97
## 75          75   Male  Engineering  26              87         56
## 76          76 Female     Business  29              58         60
## 77          77 Female     Business  28              43         50
## 78          78 Female Data Science  23              74         85
## 79          79   Male           IT  20              88         91
## 80          80 Female  Engineering  21              56         63
## 81          81   Male  Engineering  29              83         81
## 82          82 Female Data Science  28              67         96
## 83          83   Male  Engineering  21              69         50
## 84          84 Female           IT  20              68         82
## 85          85   Male Data Science  27              85         73
## 86          86   Male           IT  20              92         67
## 87          87 Female Data Science  20              45         57
## 88          88   Male Data Science  21              74         44
## 89          89 Female     Business  24              99         68
## 90          90 Female           IT  21              80         86
## 91          91 Female  Engineering  26              76         44
## 92          92 Female  Engineering  18              63         53
## 93          93 Female           IT  25              68         92
```

```
## 94          94 Female  Engineering  24                88          35
## 95          95 Female  Engineering  19                85          38
## 96          96 Female           IT  25                92          50
## 97          97 Female Data Science  18                70          58
## 98          98 Female           IT  28                74          36
## 99          99 Female Data Science  29                99          66
## 100        100   Male  Engineering  26                72          58
## 101        101   Male  Engineering  26                91          46
## 102        102 Female  Engineering  19                60          84
## 103        103 Female     Business  24                71          69
## 104        104 Female Data Science  27                62          67
## 105        105 Female     Business  20                72          67
## 106        106 Female     Business  24                42          95
## 107        107 Female           IT  29                57          85
## 108        108 Female     Business  27                64          77
## 109        109 Female           IT  26                81          46
## 110        110   Male           IT  21                70          99
## 111        111 Female     Business  29                93          67
## 112        112   Male Data Science  18                97          74
## 113        113 Female Data Science  19                42          77
## 114        114 Female Data Science  18                79          78
## 115        115   Male Data Science  29                85          63
## 116        116 Female Data Science  22                63          47
## 117        117   Male     Business  22                89          46
## 118        118 Female     Business  28                71          80
## 119        119 Female  Engineering  24                86          36
## 120        120   Male Data Science  26                61          69
##     Attendance_Percentage Total_Score Grade
## 1                      72        66.2     C
## 2                      66        66.2     C
## 3                      75        44.8     F
## 4                      57        65.4     C
## 5                      78        62.2     C
## 6                      75        42.6     F
## 7                      59        69.8     C
## 8                      75        74.4     B
## 9                      83        81.0     A
## 10                     90        83.6     A
## 11                     56        64.8     C
## 12                     53        75.6     B
## 13                     99        71.8     B
## 14                     94        54.8     F
## 15                     60        63.4     C
## 16                     78        75.2     B
## 17                     85        54.6     F
## 18                     74        85.4     A
## 19                     70        73.6     B
## 20                     85        71.0     B
## 21                     59        62.0     C
## 22                     86        94.2     A
## 23                     58        50.8     F
## 24                     73        59.6     F
## 25                     84        54.4     F
## 26                     98        56.6     F
```

```
## 27                    84        62.4     C
## 28                    97        92.2     A
## 29                    85        73.8     B
## 30                    67        51.8     F
## 31                    98        92.4     A
## 32                    88        87.8     A
## 33                    81        79.6     B
## 34                    73        65.6     C
## 35                    72        85.2     A
## 36                    81        84.0     A
## 37                    86        67.0     C
## 38                    61        80.4     A
## 39                    98        66.8     C
## 40                    62        53.4     F
## 41                    72        87.6     A
## 42                    74        80.0     A
## 43                    84        63.2     C
## 44                    90        71.0     B
## 45                    79        74.8     B
## 46                    66        40.2     F
## 47                    98        46.4     F
## 48                    69        77.4     B
## 49                    97        88.2     A
## 50                    74        64.8     C
## 51                    71        90.8     A
## 52                    62        66.0     C
## 53                    68        52.6     F
## 54                    98        76.0     B
## 55                    85        56.2     F
## 56                    61        64.6     C
## 57                    90        85.8     A
## 58                    68        62.6     C
## 59                    61        86.0     A
## 60                    58        87.0     A
## 61                    56        93.6     A
## 62                    77        71.4     B
## 63                    63        44.6     F
## 64                    80        54.0     F
## 65                    68        47.4     F
## 66                    96        84.4     A
## 67                    65        45.0     F
## 68                    54        68.4     C
## 69                    84        78.0     B
## 70                    61        47.8     F
## 71                    74        64.4     C
## 72                    70        85.8     A
## 73                    85        76.4     B
## 74                    72        97.8     A
## 75                    65        68.4     C
## 76                    88        59.2     F
## 77                    94        47.2     F
## 78                    91        80.6     A
## 79                    88        89.8     A
## 80                    63        60.2     C
```

```
## 81                    80        81.8     A
## 82                    54        84.4     A
## 83                    84        57.6     F
## 84                    72        76.4     B
## 85                    78        77.8     B
## 86                    92        77.0     B
## 87                    60        52.2     F
## 88                    67        56.0     F
## 89                    96        80.4     A
## 90                    61        83.6     A
## 91                    58        56.8     F
## 92                    59        57.0     F
## 93                    93        82.4     A
## 94                    66        56.2     F
## 95                    87        56.8     F
## 96                    56        66.8     C
## 97                    95        62.8     C
## 98                    62        51.2     F
## 99                    89        79.2     B
## 100                   91        63.6     C
## 101                   58        64.0     C
## 102                   99        74.4     B
## 103                   76        69.8     C
## 104                   51        65.0     C
## 105                   54        69.0     C
## 106                   78        73.8     B
## 107                   86        73.8     B
## 108                   87        71.8     B
## 109                   68        60.0     C
## 110                   57        87.4     A
## 111                   97        77.4     B
## 112                   94        83.2     A
## 113                   50        63.0     C
## 114                   71        78.4     B
## 115                   66        71.8     B
## 116                   56        53.4     F
## 117                   74        63.2     C
## 118                   94        76.4     B
## 119                   53        56.0     F
## 120                   85        65.8     C
```

##1 CREATE NEW COLUMN honours_category

```
# Ensure you use ~ and check that Total_Score matches your csv column name exactly
df<-df%>%
  mutate(
    Total_Score=Assignment_Score+Exam_Score, # Create the missing column
    Honours_Category=case_when(
    Total_Score>=75~"First Class",
    Total_Score>=65~"Second Class Upper",
    Total_Score>=55~"Second Class Lower ",
    Total_Score>=50~"Pass",
    TRUE            ~"Fail"
    ))
```

```r
print(df)
```

```
##     Student_ID Gender   Department Age Assignment_Score Exam_Score
## 1            1   Male Data Science  27               80         57
## 2            2 Female           IT  24               74         61
## 3            3   Male     Business  29               58         36
## 4            4   Male  Engineering  26               87         51
## 5            5   Male           IT  24               55         67
## 6            6 Female Data Science  18               42         43
## 7            7   Male     Business  18               59         77
## 8            8   Male           IT  26               63         82
## 9            9   Male Data Science  28               93         73
## 10          10 Female Data Science  26               95         76
## 11          11   Male Data Science  21               72         60
## 12          12   Male           IT  26               63         84
## 13          13   Male           IT  20               91         59
## 14          14   Male           IT  24               50         58
## 15          15 Female  Engineering  23               88         47
## 16          16   Male           IT  25               47         94
## 17          17 Female           IT  28               75         41
## 18          18 Female           IT  26               77         91
## 19          19 Female  Engineering  22               79         70
## 20          20   Male           IT  18               59         79
## 21          21 Female Data Science  20               74         54
## 22          22   Male           IT  27               87         99
## 23          23 Female Data Science  29               64         42
## 24          24 Female Data Science  25               74         50
## 25          25 Female Data Science  28               64         48
## 26          26 Female  Engineering  23               68         49
## 27          27 Female  Engineering  25               57         66
## 28          28 Female  Engineering  26               85         97
## 29          29 Female           IT  21               57         85
## 30          30 Female Data Science  18               41         59
## 31          31   Male  Engineering  18               93         92
## 32          32   Male  Engineering  27               74         97
## 33          33 Female           IT  21               55         96
## 34          34 Female  Engineering  29               80         56
## 35          35 Female           IT  24               75         92
## 36          36   Male     Business  19               72         92
## 37          37 Female  Engineering  20               43         83
## 38          38   Male     Business  18               72         86
## 39          39   Male           IT  22               53         76
## 40          40   Male Data Science  18               60         49
## 41          41   Male  Engineering  25               87         88
## 42          42   Male           IT  18               59         94
## 43          43 Female Data Science  28               95         42
## 44          44 Female Data Science  18               47         87
## 45          45 Female     Business  19               46         94
## 46          46 Female           IT  19               42         39
## 47          47 Female Data Science  29               56         40
## 48          48   Male  Engineering  23               72         81
## 49          49 Female  Engineering  24               87         89
## 50          50 Female     Business  22               51         74
```

```
## 51          51   Male Data Science  18              98         86
## 52          52 Female           IT  18              90         50
## 53          53   Male  Engineering  20              61         47
## 54          54 Female Data Science  19              94         64
## 55          55   Male Data Science  22              61         53
## 56          56 Female     Business  29              85         51
## 57          57 Female  Engineering  27              69         97
## 58          58   Male  Engineering  23              77         53
## 59          59   Male           IT  24              77         92
## 60          60   Male  Engineering  21              84         89
## 61          61   Male           IT  24              90         96
## 62          62   Male  Engineering  28              93         57
## 63          63   Male     Business  25              47         43
## 64          64   Male  Engineering  28              66         46
## 65          65   Male           IT  18              66         35
## 66          66 Female           IT  23              73         92
## 67          67 Female     Business  25              60         35
## 68          68   Male  Engineering  22              69         68
## 69          69 Female  Engineering  21              72         82
## 70          70 Female     Business  19              67         35
## 71          71 Female  Engineering  23              86         50
## 72          72 Female  Engineering  23              72         95
## 73          73   Male           IT  28              44         98
## 74          74 Female  Engineering  18              99         97
## 75          75   Male  Engineering  26              87         56
## 76          76 Female     Business  29              58         60
## 77          77 Female     Business  28              43         50
## 78          78 Female Data Science  23              74         85
## 79          79   Male           IT  20              88         91
## 80          80 Female  Engineering  21              56         63
## 81          81   Male  Engineering  29              83         81
## 82          82 Female Data Science  28              67         96
## 83          83   Male  Engineering  21              69         50
## 84          84 Female           IT  20              68         82
## 85          85   Male Data Science  27              85         73
## 86          86   Male           IT  20              92         67
## 87          87 Female Data Science  20              45         57
## 88          88   Male Data Science  21              74         44
## 89          89 Female     Business  24              99         68
## 90          90 Female           IT  21              80         86
## 91          91 Female  Engineering  26              76         44
## 92          92 Female  Engineering  18              63         53
## 93          93 Female           IT  25              68         92
## 94          94 Female  Engineering  24              88         35
## 95          95 Female  Engineering  19              85         38
## 96          96 Female           IT  25              92         50
## 97          97 Female Data Science  18              70         58
## 98          98 Female           IT  28              74         36
## 99          99 Female Data Science  29              99         66
## 100        100   Male  Engineering  26              72         58
## 101        101   Male  Engineering  26              91         46
## 102        102 Female  Engineering  19              60         84
## 103        103 Female     Business  24              71         69
## 104        104 Female Data Science  27              62         67
```

```
## 105        105 Female     Business  20               72         67
## 106        106 Female     Business  24               42         95
## 107        107 Female           IT  29               57         85
## 108        108 Female     Business  27               64         77
## 109        109 Female           IT  26               81         46
## 110        110   Male           IT  21               70         99
## 111        111 Female     Business  29               93         67
## 112        112   Male Data Science  18               97         74
## 113        113 Female Data Science  19               42         77
## 114        114 Female Data Science  18               79         78
## 115        115   Male Data Science  29               85         63
## 116        116 Female Data Science  22               63         47
## 117        117   Male     Business  22               89         46
## 118        118 Female     Business  28               71         80
## 119        119 Female  Engineering  24               86         36
## 120        120   Male Data Science  26               61         69
##     Attendance_Percentage Total_Score Grade Honours_Category
## 1                      72         137     C      First Class
## 2                      66         135     C      First Class
## 3                      75          94     F      First Class
## 4                      57         138     C      First Class
## 5                      78         122     C      First Class
## 6                      75          85     F      First Class
## 7                      59         136     C      First Class
## 8                      75         145     B      First Class
## 9                      83         166     A      First Class
## 10                     90         171     A      First Class
## 11                     56         132     C      First Class
## 12                     53         147     B      First Class
## 13                     99         150     B      First Class
## 14                     94         108     F      First Class
## 15                     60         135     C      First Class
## 16                     78         141     B      First Class
## 17                     85         116     F      First Class
## 18                     74         168     A      First Class
## 19                     70         149     B      First Class
## 20                     85         138     B      First Class
## 21                     59         128     C      First Class
## 22                     86         186     A      First Class
## 23                     58         106     F      First Class
## 24                     73         124     F      First Class
## 25                     84         112     F      First Class
## 26                     98         117     F      First Class
## 27                     84         123     C      First Class
## 28                     97         182     A      First Class
## 29                     85         142     B      First Class
## 30                     67         100     F      First Class
## 31                     98         185     A      First Class
## 32                     88         171     A      First Class
## 33                     81         151     B      First Class
## 34                     73         136     C      First Class
## 35                     72         167     A      First Class
## 36                     81         164     A      First Class
## 37                     86         126     C      First Class
```

```
## 38                    61        158     A      First Class
## 39                    98        129     C      First Class
## 40                    62        109     F      First Class
## 41                    72        175     A      First Class
## 42                    74        153     A      First Class
## 43                    84        137     C      First Class
## 44                    90        134     B      First Class
## 45                    79        140     B      First Class
## 46                    66         81     F      First Class
## 47                    98         96     F      First Class
## 48                    69        153     B      First Class
## 49                    97        176     A      First Class
## 50                    74        125     C      First Class
## 51                    71        184     A      First Class
## 52                    62        140     C      First Class
## 53                    68        108     F      First Class
## 54                    98        158     B      First Class
## 55                    85        114     F      First Class
## 56                    61        136     C      First Class
## 57                    90        166     A      First Class
## 58                    68        130     C      First Class
## 59                    61        169     A      First Class
## 60                    58        173     A      First Class
## 61                    56        186     A      First Class
## 62                    77        150     B      First Class
## 63                    63         90     F      First Class
## 64                    80        112     F      First Class
## 65                    68        101     F      First Class
## 66                    96        165     A      First Class
## 67                    65         95     F      First Class
## 68                    54        137     C      First Class
## 69                    84        154     B      First Class
## 70                    61        102     F      First Class
## 71                    74        136     C      First Class
## 72                    70        167     A      First Class
## 73                    85        142     B      First Class
## 74                    72        196     A      First Class
## 75                    65        143     C      First Class
## 76                    88        118     F      First Class
## 77                    94         93     F      First Class
## 78                    91        159     A      First Class
## 79                    88        179     A      First Class
## 80                    63        119     C      First Class
## 81                    80        164     A      First Class
## 82                    54        163     A      First Class
## 83                    84        119     F      First Class
## 84                    72        150     B      First Class
## 85                    78        158     B      First Class
## 86                    92        159     B      First Class
## 87                    60        102     F      First Class
## 88                    67        118     F      First Class
## 89                    96        167     A      First Class
## 90                    61        166     A      First Class
## 91                    58        120     F      First Class
```

```
## 92                    59         116     F      First Class
## 93                    93         160     A      First Class
## 94                    66         123     F      First Class
## 95                    87         123     F      First Class
## 96                    56         142     C      First Class
## 97                    95         128     C      First Class
## 98                    62         110     F      First Class
## 99                    89         165     B      First Class
## 100                   91         130     C      First Class
## 101                   58         137     C      First Class
## 102                   99         144     B      First Class
## 103                   76         140     C      First Class
## 104                   51         129     C      First Class
## 105                   54         139     C      First Class
## 106                   78         137     B      First Class
## 107                   86         142     B      First Class
## 108                   87         141     B      First Class
## 109                   68         127     C      First Class
## 110                   57         169     A      First Class
## 111                   97         160     B      First Class
## 112                   94         171     A      First Class
## 113                   50         119     C      First Class
## 114                   71         157     B      First Class
## 115                   66         148     B      First Class
## 116                   56         110     F      First Class
## 117                   74         135     C      First Class
## 118                   94         151     B      First Class
## 119                   53         122     F      First Class
## 120                   85         130     C      First Class
```

#2. IDENTIFY STUDENTS ELIGIBLE FOR RETAKE

```r
df<-df%>%
  mutate(Retake_Required=if_else(Total_Score<50&Attendance_Percentage>70,"Yes","No"))
print(df)
```

```
##     Student_ID Gender   Department Age Assignment_Score Exam_Score
## 1            1   Male Data Science  27               80         57
## 2            2 Female           IT  24               74         61
## 3            3   Male     Business  29               58         36
## 4            4   Male  Engineering  26               87         51
## 5            5   Male           IT  24               55         67
## 6            6 Female Data Science  18               42         43
## 7            7   Male     Business  18               59         77
## 8            8   Male           IT  26               63         82
## 9            9   Male Data Science  28               93         73
## 10          10 Female Data Science  26               95         76
## 11          11   Male Data Science  21               72         60
## 12          12   Male           IT  26               63         84
## 13          13   Male           IT  20               91         59
## 14          14   Male           IT  24               50         58
## 15          15 Female  Engineering  23               88         47
## 16          16   Male           IT  25               47         94
```

```
## 17          17 Female           IT  28               75         41
## 18          18 Female           IT  26               77         91
## 19          19 Female  Engineering  22               79         70
## 20          20   Male           IT  18               59         79
## 21          21 Female Data Science  20               74         54
## 22          22   Male           IT  27               87         99
## 23          23 Female Data Science  29               64         42
## 24          24 Female Data Science  25               74         50
## 25          25 Female Data Science  28               64         48
## 26          26 Female  Engineering  23               68         49
## 27          27 Female  Engineering  25               57         66
## 28          28 Female  Engineering  26               85         97
## 29          29 Female           IT  21               57         85
## 30          30 Female Data Science  18               41         59
## 31          31   Male  Engineering  18               93         92
## 32          32   Male  Engineering  27               74         97
## 33          33 Female           IT  21               55         96
## 34          34 Female  Engineering  29               80         56
## 35          35 Female           IT  24               75         92
## 36          36   Male     Business  19               72         92
## 37          37 Female  Engineering  20               43         83
## 38          38   Male     Business  18               72         86
## 39          39   Male           IT  22               53         76
## 40          40   Male Data Science  18               60         49
## 41          41   Male  Engineering  25               87         88
## 42          42   Male           IT  18               59         94
## 43          43 Female Data Science  28               95         42
## 44          44 Female Data Science  18               47         87
## 45          45 Female     Business  19               46         94
## 46          46 Female           IT  19               42         39
## 47          47 Female Data Science  29               56         40
## 48          48   Male  Engineering  23               72         81
## 49          49 Female  Engineering  24               87         89
## 50          50 Female     Business  22               51         74
## 51          51   Male Data Science  18               98         86
## 52          52 Female           IT  18               90         50
## 53          53   Male  Engineering  20               61         47
## 54          54 Female Data Science  19               94         64
## 55          55   Male Data Science  22               61         53
## 56          56 Female     Business  29               85         51
## 57          57 Female  Engineering  27               69         97
## 58          58   Male  Engineering  23               77         53
## 59          59   Male           IT  24               77         92
## 60          60   Male  Engineering  21               84         89
## 61          61   Male           IT  24               90         96
## 62          62   Male  Engineering  28               93         57
## 63          63   Male     Business  25               47         43
## 64          64   Male  Engineering  28               66         46
## 65          65   Male           IT  18               66         35
## 66          66 Female           IT  23               73         92
## 67          67 Female     Business  25               60         35
## 68          68   Male  Engineering  22               69         68
## 69          69 Female  Engineering  21               72         82
## 70          70 Female     Business  19               67         35
```

```
## 71          71 Female  Engineering  23               86         50
## 72          72 Female  Engineering  23               72         95
## 73          73   Male           IT  28               44         98
## 74          74 Female  Engineering  18               99         97
## 75          75   Male  Engineering  26               87         56
## 76          76 Female     Business  29               58         60
## 77          77 Female     Business  28               43         50
## 78          78 Female Data Science  23               74         85
## 79          79   Male           IT  20               88         91
## 80          80 Female  Engineering  21               56         63
## 81          81   Male  Engineering  29               83         81
## 82          82 Female Data Science  28               67         96
## 83          83   Male  Engineering  21               69         50
## 84          84 Female           IT  20               68         82
## 85          85   Male Data Science  27               85         73
## 86          86   Male           IT  20               92         67
## 87          87 Female Data Science  20               45         57
## 88          88   Male Data Science  21               74         44
## 89          89 Female     Business  24               99         68
## 90          90 Female           IT  21               80         86
## 91          91 Female  Engineering  26               76         44
## 92          92 Female  Engineering  18               63         53
## 93          93 Female           IT  25               68         92
## 94          94 Female  Engineering  24               88         35
## 95          95 Female  Engineering  19               85         38
## 96          96 Female           IT  25               92         50
## 97          97 Female Data Science  18               70         58
## 98          98 Female           IT  28               74         36
## 99          99 Female Data Science  29               99         66
## 100        100   Male  Engineering  26               72         58
## 101        101   Male  Engineering  26               91         46
## 102        102 Female  Engineering  19               60         84
## 103        103 Female     Business  24               71         69
## 104        104 Female Data Science  27               62         67
## 105        105 Female     Business  20               72         67
## 106        106 Female     Business  24               42         95
## 107        107 Female           IT  29               57         85
## 108        108 Female     Business  27               64         77
## 109        109 Female           IT  26               81         46
## 110        110   Male           IT  21               70         99
## 111        111 Female     Business  29               93         67
## 112        112   Male Data Science  18               97         74
## 113        113 Female Data Science  19               42         77
## 114        114 Female Data Science  18               79         78
## 115        115   Male Data Science  29               85         63
## 116        116 Female Data Science  22               63         47
## 117        117   Male     Business  22               89         46
## 118        118 Female     Business  28               71         80
## 119        119 Female  Engineering  24               86         36
## 120        120   Male Data Science  26               61         69
##     Attendance_Percentage Total_Score Grade Honours_Category Retake_Required
## 1                      72         137     C      First Class              No
## 2                      66         135     C      First Class              No
## 3                      75          94     F      First Class              No
```

```
## 4                    57        138    C     First Class          No
## 5                    78        122    C     First Class          No
## 6                    75         85    F     First Class          No
## 7                    59        136    C     First Class          No
## 8                    75        145    B     First Class          No
## 9                    83        166    A     First Class          No
## 10                   90        171    A     First Class          No
## 11                   56        132    C     First Class          No
## 12                   53        147    B     First Class          No
## 13                   99        150    B     First Class          No
## 14                   94        108    F     First Class          No
## 15                   60        135    C     First Class          No
## 16                   78        141    B     First Class          No
## 17                   85        116    F     First Class          No
## 18                   74        168    A     First Class          No
## 19                   70        149    B     First Class          No
## 20                   85        138    B     First Class          No
## 21                   59        128    C     First Class          No
## 22                   86        186    A     First Class          No
## 23                   58        106    F     First Class          No
## 24                   73        124    F     First Class          No
## 25                   84        112    F     First Class          No
## 26                   98        117    F     First Class          No
## 27                   84        123    C     First Class          No
## 28                   97        182    A     First Class          No
## 29                   85        142    B     First Class          No
## 30                   67        100    F     First Class          No
## 31                   98        185    A     First Class          No
## 32                   88        171    A     First Class          No
## 33                   81        151    B     First Class          No
## 34                   73        136    C     First Class          No
## 35                   72        167    A     First Class          No
## 36                   81        164    A     First Class          No
## 37                   86        126    C     First Class          No
## 38                   61        158    A     First Class          No
## 39                   98        129    C     First Class          No
## 40                   62        109    F     First Class          No
## 41                   72        175    A     First Class          No
## 42                   74        153    A     First Class          No
## 43                   84        137    C     First Class          No
## 44                   90        134    B     First Class          No
## 45                   79        140    B     First Class          No
## 46                   66         81    F     First Class          No
## 47                   98         96    F     First Class          No
## 48                   69        153    B     First Class          No
## 49                   97        176    A     First Class          No
## 50                   74        125    C     First Class          No
## 51                   71        184    A     First Class          No
## 52                   62        140    C     First Class          No
## 53                   68        108    F     First Class          No
## 54                   98        158    B     First Class          No
## 55                   85        114    F     First Class          No
## 56                   61        136    C     First Class          No
## 57                   90        166    A     First Class          No
```

```
## 58                    68        130    C     First Class            No
## 59                    61        169    A     First Class            No
## 60                    58        173    A     First Class            No
## 61                    56        186    A     First Class            No
## 62                    77        150    B     First Class            No
## 63                    63         90    F     First Class            No
## 64                    80        112    F     First Class            No
## 65                    68        101    F     First Class            No
## 66                    96        165    A     First Class            No
## 67                    65         95    F     First Class            No
## 68                    54        137    C     First Class            No
## 69                    84        154    B     First Class            No
## 70                    61        102    F     First Class            No
## 71                    74        136    C     First Class            No
## 72                    70        167    A     First Class            No
## 73                    85        142    B     First Class            No
## 74                    72        196    A     First Class            No
## 75                    65        143    C     First Class            No
## 76                    88        118    F     First Class            No
## 77                    94         93    F     First Class            No
## 78                    91        159    A     First Class            No
## 79                    88        179    A     First Class            No
## 80                    63        119    C     First Class            No
## 81                    80        164    A     First Class            No
## 82                    54        163    A     First Class            No
## 83                    84        119    F     First Class            No
## 84                    72        150    B     First Class            No
## 85                    78        158    B     First Class            No
## 86                    92        159    B     First Class            No
## 87                    60        102    F     First Class            No
## 88                    67        118    F     First Class            No
## 89                    96        167    A     First Class            No
## 90                    61        166    A     First Class            No
## 91                    58        120    F     First Class            No
## 92                    59        116    F     First Class            No
## 93                    93        160    A     First Class            No
## 94                    66        123    F     First Class            No
## 95                    87        123    F     First Class            No
## 96                    56        142    C     First Class            No
## 97                    95        128    C     First Class            No
## 98                    62        110    F     First Class            No
## 99                    89        165    B     First Class            No
## 100                   91        130    C     First Class            No
## 101                   58        137    C     First Class            No
## 102                   99        144    B     First Class            No
## 103                   76        140    C     First Class            No
## 104                   51        129    C     First Class            No
## 105                   54        139    C     First Class            No
## 106                   78        137    B     First Class            No
## 107                   86        142    B     First Class            No
## 108                   87        141    B     First Class            No
## 109                   68        127    C     First Class            No
## 110                   57        169    A     First Class            No
## 111                   97        160    B     First Class            No
```

```
## 112                    94         171     A      First Class              No
## 113                    50         119     C      First Class              No
## 114                    71         157     B      First Class              No
## 115                    66         148     B      First Class              No
## 116                    56         110     F      First Class              No
## 117                    74         135     C      First Class              No
## 118                    94         151     B      First Class              No
## 119                    53         122     F      First Class              No
## 120                    85         130     C      First Class              No
```

#3.IDENTIFY YOUNG HIGH PERFOMERS

```r
# THIS CREATES A NEW DATAFRAME CONTAINING ONLY THE STUDENT WHO MEET BOTH CRITERIA
young_high_perfomers<-df %>%
  filter(Age<=22, Total_Score>=70)
print(young_high_perfomers)
```

```
##    Student_ID Gender   Department Age Assignment_Score Exam_Score
## 1           6 Female Data Science  18               42         43
## 2           7   Male     Business  18               59         77
## 3          11   Male Data Science  21               72         60
## 4          13   Male           IT  20               91         59
## 5          19 Female  Engineering  22               79         70
## 6          20   Male           IT  18               59         79
## 7          21 Female Data Science  20               74         54
## 8          29 Female           IT  21               57         85
## 9          30 Female Data Science  18               41         59
## 10         31   Male  Engineering  18               93         92
## 11         33 Female           IT  21               55         96
## 12         36   Male     Business  19               72         92
## 13         37 Female  Engineering  20               43         83
## 14         38   Male     Business  18               72         86
## 15         39   Male           IT  22               53         76
## 16         40   Male Data Science  18               60         49
## 17         42   Male           IT  18               59         94
## 18         44 Female Data Science  18               47         87
## 19         45 Female     Business  19               46         94
## 20         46 Female           IT  19               42         39
## 21         50 Female     Business  22               51         74
## 22         51   Male Data Science  18               98         86
## 23         52 Female           IT  18               90         50
## 24         53   Male  Engineering  20               61         47
## 25         54 Female Data Science  19               94         64
## 26         55   Male Data Science  22               61         53
## 27         60   Male  Engineering  21               84         89
## 28         65   Male           IT  18               66         35
## 29         68   Male  Engineering  22               69         68
## 30         69 Female  Engineering  21               72         82
## 31         70 Female     Business  19               67         35
## 32         74 Female  Engineering  18               99         97
## 33         79   Male           IT  20               88         91
## 34         80 Female  Engineering  21               56         63
## 35         83   Male  Engineering  21               69         50
```

```
## 36         84 Female           IT  20               68         82
## 37         86   Male           IT  20               92         67
## 38         87 Female Data Science  20               45         57
## 39         88   Male Data Science  21               74         44
## 40         90 Female           IT  21               80         86
## 41         92 Female  Engineering  18               63         53
## 42         95 Female  Engineering  19               85         38
## 43         97 Female Data Science  18               70         58
## 44        102 Female  Engineering  19               60         84
## 45        105 Female     Business  20               72         67
## 46        110   Male           IT  21               70         99
## 47        112   Male Data Science  18               97         74
## 48        113 Female Data Science  19               42         77
## 49        114 Female Data Science  18               79         78
## 50        116 Female Data Science  22               63         47
## 51        117   Male     Business  22               89         46
##    Attendance_Percentage Total_Score Grade Honours_Category Retake_Required
## 1                     75          85     F      First Class              No
## 2                     59         136     C      First Class              No
## 3                     56         132     C      First Class              No
## 4                     99         150     B      First Class              No
## 5                     70         149     B      First Class              No
## 6                     85         138     B      First Class              No
## 7                     59         128     C      First Class              No
## 8                     85         142     B      First Class              No
## 9                     67         100     F      First Class              No
## 10                    98         185     A      First Class              No
## 11                    81         151     B      First Class              No
## 12                    81         164     A      First Class              No
## 13                    86         126     C      First Class              No
## 14                    61         158     A      First Class              No
## 15                    98         129     C      First Class              No
## 16                    62         109     F      First Class              No
## 17                    74         153     A      First Class              No
## 18                    90         134     B      First Class              No
## 19                    79         140     B      First Class              No
## 20                    66          81     F      First Class              No
## 21                    74         125     C      First Class              No
## 22                    71         184     A      First Class              No
## 23                    62         140     C      First Class              No
## 24                    68         108     F      First Class              No
## 25                    98         158     B      First Class              No
## 26                    85         114     F      First Class              No
## 27                    58         173     A      First Class              No
## 28                    68         101     F      First Class              No
## 29                    54         137     C      First Class              No
## 30                    84         154     B      First Class              No
## 31                    61         102     F      First Class              No
## 32                    72         196     A      First Class              No
## 33                    88         179     A      First Class              No
## 34                    63         119     C      First Class              No
## 35                    84         119     F      First Class              No
## 36                    72         150     B      First Class              No
## 37                    92         159     B      First Class              No
```

```
## 38                    60         102     F      First Class              No
## 39                    67         118     F      First Class              No
## 40                    61         166     A      First Class              No
## 41                    59         116     F      First Class              No
## 42                    87         123     F      First Class              No
## 43                    95         128     C      First Class              No
## 44                    99         144     B      First Class              No
## 45                    54         139     C      First Class              No
## 46                    57         169     A      First Class              No
## 47                    94         171     A      First Class              No
## 48                    50         119     C      First Class              No
## 49                    71         157     B      First Class              No
## 50                    56         110     F      First Class              No
## 51                    74         135     C      First Class              No
```

#4.ATTENDANCE WARNING CATEGORY

```r
df <- df %>%
  mutate(Attendance_Status = case_when(
    Attendance_Percentage < 60 ~ "Critical",
    Attendance_Percentage >= 60 & Attendance_Percentage <= 74 ~ "Warning",
    Attendance_Percentage >= 75 ~ "Good Standing",
    TRUE ~ "Unknown"
  ))
print(df)
```

```
##     Student_ID Gender   Department Age Assignment_Score Exam_Score
## 1            1   Male Data Science  27               80         57
## 2            2 Female           IT  24               74         61
## 3            3   Male     Business  29               58         36
## 4            4   Male  Engineering  26               87         51
## 5            5   Male           IT  24               55         67
## 6            6 Female Data Science  18               42         43
## 7            7   Male     Business  18               59         77
## 8            8   Male           IT  26               63         82
## 9            9   Male Data Science  28               93         73
## 10          10 Female Data Science  26               95         76
## 11          11   Male Data Science  21               72         60
## 12          12   Male           IT  26               63         84
## 13          13   Male           IT  20               91         59
## 14          14   Male           IT  24               50         58
## 15          15 Female  Engineering  23               88         47
## 16          16   Male           IT  25               47         94
## 17          17 Female           IT  28               75         41
## 18          18 Female           IT  26               77         91
## 19          19 Female  Engineering  22               79         70
## 20          20   Male           IT  18               59         79
## 21          21 Female Data Science  20               74         54
## 22          22   Male           IT  27               87         99
## 23          23 Female Data Science  29               64         42
## 24          24 Female Data Science  25               74         50
## 25          25 Female Data Science  28               64         48
## 26          26 Female  Engineering  23               68         49
```

```
## 27          27 Female  Engineering  25              57         66
## 28          28 Female  Engineering  26              85         97
## 29          29 Female           IT  21              57         85
## 30          30 Female Data Science  18              41         59
## 31          31   Male  Engineering  18              93         92
## 32          32   Male  Engineering  27              74         97
## 33          33 Female           IT  21              55         96
## 34          34 Female  Engineering  29              80         56
## 35          35 Female           IT  24              75         92
## 36          36   Male     Business  19              72         92
## 37          37 Female  Engineering  20              43         83
## 38          38   Male     Business  18              72         86
## 39          39   Male           IT  22              53         76
## 40          40   Male Data Science  18              60         49
## 41          41   Male  Engineering  25              87         88
## 42          42   Male           IT  18              59         94
## 43          43 Female Data Science  28              95         42
## 44          44 Female Data Science  18              47         87
## 45          45 Female     Business  19              46         94
## 46          46 Female           IT  19              42         39
## 47          47 Female Data Science  29              56         40
## 48          48   Male  Engineering  23              72         81
## 49          49 Female  Engineering  24              87         89
## 50          50 Female     Business  22              51         74
## 51          51   Male Data Science  18              98         86
## 52          52 Female           IT  18              90         50
## 53          53   Male  Engineering  20              61         47
## 54          54 Female Data Science  19              94         64
## 55          55   Male Data Science  22              61         53
## 56          56 Female     Business  29              85         51
## 57          57 Female  Engineering  27              69         97
## 58          58   Male  Engineering  23              77         53
## 59          59   Male           IT  24              77         92
## 60          60   Male  Engineering  21              84         89
## 61          61   Male           IT  24              90         96
## 62          62   Male  Engineering  28              93         57
## 63          63   Male     Business  25              47         43
## 64          64   Male  Engineering  28              66         46
## 65          65   Male           IT  18              66         35
## 66          66 Female           IT  23              73         92
## 67          67 Female     Business  25              60         35
## 68          68   Male  Engineering  22              69         68
## 69          69 Female  Engineering  21              72         82
## 70          70 Female     Business  19              67         35
## 71          71 Female  Engineering  23              86         50
## 72          72 Female  Engineering  23              72         95
## 73          73   Male           IT  28              44         98
## 74          74 Female  Engineering  18              99         97
## 75          75   Male  Engineering  26              87         56
## 76          76 Female     Business  29              58         60
## 77          77 Female     Business  28              43         50
## 78          78 Female Data Science  23              74         85
## 79          79   Male           IT  20              88         91
## 80          80 Female  Engineering  21              56         63
```

```
## 81          81   Male  Engineering  29               83         81
## 82          82 Female Data Science  28               67         96
## 83          83   Male  Engineering  21               69         50
## 84          84 Female           IT  20               68         82
## 85          85   Male Data Science  27               85         73
## 86          86   Male           IT  20               92         67
## 87          87 Female Data Science  20               45         57
## 88          88   Male Data Science  21               74         44
## 89          89 Female     Business  24               99         68
## 90          90 Female           IT  21               80         86
## 91          91 Female  Engineering  26               76         44
## 92          92 Female  Engineering  18               63         53
## 93          93 Female           IT  25               68         92
## 94          94 Female  Engineering  24               88         35
## 95          95 Female  Engineering  19               85         38
## 96          96 Female           IT  25               92         50
## 97          97 Female Data Science  18               70         58
## 98          98 Female           IT  28               74         36
## 99          99 Female Data Science  29               99         66
## 100        100   Male  Engineering  26               72         58
## 101        101   Male  Engineering  26               91         46
## 102        102 Female  Engineering  19               60         84
## 103        103 Female     Business  24               71         69
## 104        104 Female Data Science  27               62         67
## 105        105 Female     Business  20               72         67
## 106        106 Female     Business  24               42         95
## 107        107 Female           IT  29               57         85
## 108        108 Female     Business  27               64         77
## 109        109 Female           IT  26               81         46
## 110        110   Male           IT  21               70         99
## 111        111 Female     Business  29               93         67
## 112        112   Male Data Science  18               97         74
## 113        113 Female Data Science  19               42         77
## 114        114 Female Data Science  18               79         78
## 115        115   Male Data Science  29               85         63
## 116        116 Female Data Science  22               63         47
## 117        117   Male     Business  22               89         46
## 118        118 Female     Business  28               71         80
## 119        119 Female  Engineering  24               86         36
## 120        120   Male Data Science  26               61         69
##     Attendance_Percentage Total_Score Grade Honours_Category Retake_Required
## 1                      72         137     C      First Class              No
## 2                      66         135     C      First Class              No
## 3                      75          94     F      First Class              No
## 4                      57         138     C      First Class              No
## 5                      78         122     C      First Class              No
## 6                      75          85     F      First Class              No
## 7                      59         136     C      First Class              No
## 8                      75         145     B      First Class              No
## 9                      83         166     A      First Class              No
## 10                     90         171     A      First Class              No
## 11                     56         132     C      First Class              No
## 12                     53         147     B      First Class              No
## 13                     99         150     B      First Class              No
```

```
## 14                    94        108    F     First Class            No
## 15                    60        135    C     First Class            No
## 16                    78        141    B     First Class            No
## 17                    85        116    F     First Class            No
## 18                    74        168    A     First Class            No
## 19                    70        149    B     First Class            No
## 20                    85        138    B     First Class            No
## 21                    59        128    C     First Class            No
## 22                    86        186    A     First Class            No
## 23                    58        106    F     First Class            No
## 24                    73        124    F     First Class            No
## 25                    84        112    F     First Class            No
## 26                    98        117    F     First Class            No
## 27                    84        123    C     First Class            No
## 28                    97        182    A     First Class            No
## 29                    85        142    B     First Class            No
## 30                    67        100    F     First Class            No
## 31                    98        185    A     First Class            No
## 32                    88        171    A     First Class            No
## 33                    81        151    B     First Class            No
## 34                    73        136    C     First Class            No
## 35                    72        167    A     First Class            No
## 36                    81        164    A     First Class            No
## 37                    86        126    C     First Class            No
## 38                    61        158    A     First Class            No
## 39                    98        129    C     First Class            No
## 40                    62        109    F     First Class            No
## 41                    72        175    A     First Class            No
## 42                    74        153    A     First Class            No
## 43                    84        137    C     First Class            No
## 44                    90        134    B     First Class            No
## 45                    79        140    B     First Class            No
## 46                    66         81    F     First Class            No
## 47                    98         96    F     First Class            No
## 48                    69        153    B     First Class            No
## 49                    97        176    A     First Class            No
## 50                    74        125    C     First Class            No
## 51                    71        184    A     First Class            No
## 52                    62        140    C     First Class            No
## 53                    68        108    F     First Class            No
## 54                    98        158    B     First Class            No
## 55                    85        114    F     First Class            No
## 56                    61        136    C     First Class            No
## 57                    90        166    A     First Class            No
## 58                    68        130    C     First Class            No
## 59                    61        169    A     First Class            No
## 60                    58        173    A     First Class            No
## 61                    56        186    A     First Class            No
## 62                    77        150    B     First Class            No
## 63                    63         90    F     First Class            No
## 64                    80        112    F     First Class            No
## 65                    68        101    F     First Class            No
## 66                    96        165    A     First Class            No
## 67                    65         95    F     First Class            No
```

```
## 68                    54        137     C      First Class              No
## 69                    84        154     B      First Class              No
## 70                    61        102     F      First Class              No
## 71                    74        136     C      First Class              No
## 72                    70        167     A      First Class              No
## 73                    85        142     B      First Class              No
## 74                    72        196     A      First Class              No
## 75                    65        143     C      First Class              No
## 76                    88        118     F      First Class              No
## 77                    94         93     F      First Class              No
## 78                    91        159     A      First Class              No
## 79                    88        179     A      First Class              No
## 80                    63        119     C      First Class              No
## 81                    80        164     A      First Class              No
## 82                    54        163     A      First Class              No
## 83                    84        119     F      First Class              No
## 84                    72        150     B      First Class              No
## 85                    78        158     B      First Class              No
## 86                    92        159     B      First Class              No
## 87                    60        102     F      First Class              No
## 88                    67        118     F      First Class              No
## 89                    96        167     A      First Class              No
## 90                    61        166     A      First Class              No
## 91                    58        120     F      First Class              No
## 92                    59        116     F      First Class              No
## 93                    93        160     A      First Class              No
## 94                    66        123     F      First Class              No
## 95                    87        123     F      First Class              No
## 96                    56        142     C      First Class              No
## 97                    95        128     C      First Class              No
## 98                    62        110     F      First Class              No
## 99                    89        165     B      First Class              No
## 100                   91        130     C      First Class              No
## 101                   58        137     C      First Class              No
## 102                   99        144     B      First Class              No
## 103                   76        140     C      First Class              No
## 104                   51        129     C      First Class              No
## 105                   54        139     C      First Class              No
## 106                   78        137     B      First Class              No
## 107                   86        142     B      First Class              No
## 108                   87        141     B      First Class              No
## 109                   68        127     C      First Class              No
## 110                   57        169     A      First Class              No
## 111                   97        160     B      First Class              No
## 112                   94        171     A      First Class              No
## 113                   50        119     C      First Class              No
## 114                   71        157     B      First Class              No
## 115                   66        148     B      First Class              No
## 116                   56        110     F      First Class              No
## 117                   74        135     C      First Class              No
## 118                   94        151     B      First Class              No
## 119                   53        122     F      First Class              No
## 120                   85        130     C      First Class              No
##     Attendance_Status
```

```
## 1             Warning
## 2             Warning
## 3       Good Standing
## 4            Critical
## 5       Good Standing
## 6       Good Standing
## 7            Critical
## 8       Good Standing
## 9       Good Standing
## 10      Good Standing
## 11           Critical
## 12           Critical
## 13      Good Standing
## 14      Good Standing
## 15            Warning
## 16      Good Standing
## 17      Good Standing
## 18            Warning
## 19            Warning
## 20      Good Standing
## 21           Critical
## 22      Good Standing
## 23           Critical
## 24            Warning
## 25      Good Standing
## 26      Good Standing
## 27      Good Standing
## 28      Good Standing
## 29      Good Standing
## 30            Warning
## 31      Good Standing
## 32      Good Standing
## 33      Good Standing
## 34            Warning
## 35            Warning
## 36      Good Standing
## 37      Good Standing
## 38            Warning
## 39      Good Standing
## 40            Warning
## 41            Warning
## 42            Warning
## 43      Good Standing
## 44      Good Standing
## 45      Good Standing
## 46            Warning
## 47      Good Standing
## 48            Warning
## 49      Good Standing
## 50            Warning
## 51            Warning
## 52            Warning
## 53            Warning
## 54      Good Standing
```

```
## 55      Good Standing
## 56            Warning
## 57      Good Standing
## 58            Warning
## 59            Warning
## 60           Critical
## 61           Critical
## 62      Good Standing
## 63            Warning
## 64      Good Standing
## 65            Warning
## 66      Good Standing
## 67            Warning
## 68           Critical
## 69      Good Standing
## 70            Warning
## 71            Warning
## 72            Warning
## 73      Good Standing
## 74            Warning
## 75            Warning
## 76      Good Standing
## 77      Good Standing
## 78      Good Standing
## 79      Good Standing
## 80            Warning
## 81      Good Standing
## 82           Critical
## 83      Good Standing
## 84            Warning
## 85      Good Standing
## 86      Good Standing
## 87            Warning
## 88            Warning
## 89      Good Standing
## 90            Warning
## 91           Critical
## 92           Critical
## 93      Good Standing
## 94            Warning
## 95      Good Standing
## 96           Critical
## 97      Good Standing
## 98            Warning
## 99      Good Standing
## 100     Good Standing
## 101          Critical
## 102     Good Standing
## 103     Good Standing
## 104          Critical
## 105          Critical
## 106     Good Standing
## 107     Good Standing
## 108     Good Standing
```

```
## 109           Warning
## 110          Critical
## 111     Good Standing
## 112     Good Standing
## 113          Critical
## 114           Warning
## 115           Warning
## 116          Critical
## 117           Warning
## 118     Good Standing
## 119          Critical
## 120     Good Standing
```

#5 EXTRA CREDIT ELIGIBILITY

```
df<-df%>%
  mutate(Extra_Credit=if_else(Assignment_Score>85&Exam_Score>80,"Eligible","Ineligible"))
print(df)
```

```
##     Student_ID Gender   Department Age Assignment_Score Exam_Score
## 1            1   Male Data Science  27               80         57
## 2            2 Female           IT  24               74         61
## 3            3   Male     Business  29               58         36
## 4            4   Male  Engineering  26               87         51
## 5            5   Male           IT  24               55         67
## 6            6 Female Data Science  18               42         43
## 7            7   Male     Business  18               59         77
## 8            8   Male           IT  26               63         82
## 9            9   Male Data Science  28               93         73
## 10          10 Female Data Science  26               95         76
## 11          11   Male Data Science  21               72         60
## 12          12   Male           IT  26               63         84
## 13          13   Male           IT  20               91         59
## 14          14   Male           IT  24               50         58
## 15          15 Female  Engineering  23               88         47
## 16          16   Male           IT  25               47         94
## 17          17 Female           IT  28               75         41
## 18          18 Female           IT  26               77         91
## 19          19 Female  Engineering  22               79         70
## 20          20   Male           IT  18               59         79
## 21          21 Female Data Science  20               74         54
## 22          22   Male           IT  27               87         99
## 23          23 Female Data Science  29               64         42
## 24          24 Female Data Science  25               74         50
## 25          25 Female Data Science  28               64         48
## 26          26 Female  Engineering  23               68         49
## 27          27 Female  Engineering  25               57         66
## 28          28 Female  Engineering  26               85         97
## 29          29 Female           IT  21               57         85
## 30          30 Female Data Science  18               41         59
## 31          31   Male  Engineering  18               93         92
## 32          32   Male  Engineering  27               74         97
## 33          33 Female           IT  21               55         96
```

```
## 34          34 Female  Engineering  29              80         56
## 35          35 Female           IT  24              75         92
## 36          36   Male     Business  19              72         92
## 37          37 Female  Engineering  20              43         83
## 38          38   Male     Business  18              72         86
## 39          39   Male           IT  22              53         76
## 40          40   Male Data Science  18              60         49
## 41          41   Male  Engineering  25              87         88
## 42          42   Male           IT  18              59         94
## 43          43 Female Data Science  28              95         42
## 44          44 Female Data Science  18              47         87
## 45          45 Female     Business  19              46         94
## 46          46 Female           IT  19              42         39
## 47          47 Female Data Science  29              56         40
## 48          48   Male  Engineering  23              72         81
## 49          49 Female  Engineering  24              87         89
## 50          50 Female     Business  22              51         74
## 51          51   Male Data Science  18              98         86
## 52          52 Female           IT  18              90         50
## 53          53   Male  Engineering  20              61         47
## 54          54 Female Data Science  19              94         64
## 55          55   Male Data Science  22              61         53
## 56          56 Female     Business  29              85         51
## 57          57 Female  Engineering  27              69         97
## 58          58   Male  Engineering  23              77         53
## 59          59   Male           IT  24              77         92
## 60          60   Male  Engineering  21              84         89
## 61          61   Male           IT  24              90         96
## 62          62   Male  Engineering  28              93         57
## 63          63   Male     Business  25              47         43
## 64          64   Male  Engineering  28              66         46
## 65          65   Male           IT  18              66         35
## 66          66 Female           IT  23              73         92
## 67          67 Female     Business  25              60         35
## 68          68   Male  Engineering  22              69         68
## 69          69 Female  Engineering  21              72         82
## 70          70 Female     Business  19              67         35
## 71          71 Female  Engineering  23              86         50
## 72          72 Female  Engineering  23              72         95
## 73          73   Male           IT  28              44         98
## 74          74 Female  Engineering  18              99         97
## 75          75   Male  Engineering  26              87         56
## 76          76 Female     Business  29              58         60
## 77          77 Female     Business  28              43         50
## 78          78 Female Data Science  23              74         85
## 79          79   Male           IT  20              88         91
## 80          80 Female  Engineering  21              56         63
## 81          81   Male  Engineering  29              83         81
## 82          82 Female Data Science  28              67         96
## 83          83   Male  Engineering  21              69         50
## 84          84 Female           IT  20              68         82
## 85          85   Male Data Science  27              85         73
## 86          86   Male           IT  20              92         67
## 87          87 Female Data Science  20              45         57
```

```
## 88          88   Male Data Science  21               74         44
## 89          89 Female     Business  24               99         68
## 90          90 Female           IT  21               80         86
## 91          91 Female  Engineering  26               76         44
## 92          92 Female  Engineering  18               63         53
## 93          93 Female           IT  25               68         92
## 94          94 Female  Engineering  24               88         35
## 95          95 Female  Engineering  19               85         38
## 96          96 Female           IT  25               92         50
## 97          97 Female Data Science  18               70         58
## 98          98 Female           IT  28               74         36
## 99          99 Female Data Science  29               99         66
## 100        100   Male  Engineering  26               72         58
## 101        101   Male  Engineering  26               91         46
## 102        102 Female  Engineering  19               60         84
## 103        103 Female     Business  24               71         69
## 104        104 Female Data Science  27               62         67
## 105        105 Female     Business  20               72         67
## 106        106 Female     Business  24               42         95
## 107        107 Female           IT  29               57         85
## 108        108 Female     Business  27               64         77
## 109        109 Female           IT  26               81         46
## 110        110   Male           IT  21               70         99
## 111        111 Female     Business  29               93         67
## 112        112   Male Data Science  18               97         74
## 113        113 Female Data Science  19               42         77
## 114        114 Female Data Science  18               79         78
## 115        115   Male Data Science  29               85         63
## 116        116 Female Data Science  22               63         47
## 117        117   Male     Business  22               89         46
## 118        118 Female     Business  28               71         80
## 119        119 Female  Engineering  24               86         36
## 120        120   Male Data Science  26               61         69
##     Attendance_Percentage Total_Score Grade Honours_Category Retake_Required
## 1                      72         137     C      First Class              No
## 2                      66         135     C      First Class              No
## 3                      75          94     F      First Class              No
## 4                      57         138     C      First Class              No
## 5                      78         122     C      First Class              No
## 6                      75          85     F      First Class              No
## 7                      59         136     C      First Class              No
## 8                      75         145     B      First Class              No
## 9                      83         166     A      First Class              No
## 10                     90         171     A      First Class              No
## 11                     56         132     C      First Class              No
## 12                     53         147     B      First Class              No
## 13                     99         150     B      First Class              No
## 14                     94         108     F      First Class              No
## 15                     60         135     C      First Class              No
## 16                     78         141     B      First Class              No
## 17                     85         116     F      First Class              No
## 18                     74         168     A      First Class              No
## 19                     70         149     B      First Class              No
## 20                     85         138     B      First Class              No
```

```
## 21                    59        128    C     First Class            No
## 22                    86        186    A     First Class            No
## 23                    58        106    F     First Class            No
## 24                    73        124    F     First Class            No
## 25                    84        112    F     First Class            No
## 26                    98        117    F     First Class            No
## 27                    84        123    C     First Class            No
## 28                    97        182    A     First Class            No
## 29                    85        142    B     First Class            No
## 30                    67        100    F     First Class            No
## 31                    98        185    A     First Class            No
## 32                    88        171    A     First Class            No
## 33                    81        151    B     First Class            No
## 34                    73        136    C     First Class            No
## 35                    72        167    A     First Class            No
## 36                    81        164    A     First Class            No
## 37                    86        126    C     First Class            No
## 38                    61        158    A     First Class            No
## 39                    98        129    C     First Class            No
## 40                    62        109    F     First Class            No
## 41                    72        175    A     First Class            No
## 42                    74        153    A     First Class            No
## 43                    84        137    C     First Class            No
## 44                    90        134    B     First Class            No
## 45                    79        140    B     First Class            No
## 46                    66         81    F     First Class            No
## 47                    98         96    F     First Class            No
## 48                    69        153    B     First Class            No
## 49                    97        176    A     First Class            No
## 50                    74        125    C     First Class            No
## 51                    71        184    A     First Class            No
## 52                    62        140    C     First Class            No
## 53                    68        108    F     First Class            No
## 54                    98        158    B     First Class            No
## 55                    85        114    F     First Class            No
## 56                    61        136    C     First Class            No
## 57                    90        166    A     First Class            No
## 58                    68        130    C     First Class            No
## 59                    61        169    A     First Class            No
## 60                    58        173    A     First Class            No
## 61                    56        186    A     First Class            No
## 62                    77        150    B     First Class            No
## 63                    63         90    F     First Class            No
## 64                    80        112    F     First Class            No
## 65                    68        101    F     First Class            No
## 66                    96        165    A     First Class            No
## 67                    65         95    F     First Class            No
## 68                    54        137    C     First Class            No
## 69                    84        154    B     First Class            No
## 70                    61        102    F     First Class            No
## 71                    74        136    C     First Class            No
## 72                    70        167    A     First Class            No
## 73                    85        142    B     First Class            No
## 74                    72        196    A     First Class            No
```

```
## 75                    65         143     C      First Class              No
## 76                    88         118     F      First Class              No
## 77                    94          93     F      First Class              No
## 78                    91         159     A      First Class              No
## 79                    88         179     A      First Class              No
## 80                    63         119     C      First Class              No
## 81                    80         164     A      First Class              No
## 82                    54         163     A      First Class              No
## 83                    84         119     F      First Class              No
## 84                    72         150     B      First Class              No
## 85                    78         158     B      First Class              No
## 86                    92         159     B      First Class              No
## 87                    60         102     F      First Class              No
## 88                    67         118     F      First Class              No
## 89                    96         167     A      First Class              No
## 90                    61         166     A      First Class              No
## 91                    58         120     F      First Class              No
## 92                    59         116     F      First Class              No
## 93                    93         160     A      First Class              No
## 94                    66         123     F      First Class              No
## 95                    87         123     F      First Class              No
## 96                    56         142     C      First Class              No
## 97                    95         128     C      First Class              No
## 98                    62         110     F      First Class              No
## 99                    89         165     B      First Class              No
## 100                   91         130     C      First Class              No
## 101                   58         137     C      First Class              No
## 102                   99         144     B      First Class              No
## 103                   76         140     C      First Class              No
## 104                   51         129     C      First Class              No
## 105                   54         139     C      First Class              No
## 106                   78         137     B      First Class              No
## 107                   86         142     B      First Class              No
## 108                   87         141     B      First Class              No
## 109                   68         127     C      First Class              No
## 110                   57         169     A      First Class              No
## 111                   97         160     B      First Class              No
## 112                   94         171     A      First Class              No
## 113                   50         119     C      First Class              No
## 114                   71         157     B      First Class              No
## 115                   66         148     B      First Class              No
## 116                   56         110     F      First Class              No
## 117                   74         135     C      First Class              No
## 118                   94         151     B      First Class              No
## 119                   53         122     F      First Class              No
## 120                   85         130     C      First Class              No
##     Attendance_Status Extra_Credit
## 1             Warning   Ineligible
## 2             Warning   Ineligible
## 3       Good Standing   Ineligible
## 4            Critical   Ineligible
## 5       Good Standing   Ineligible
## 6       Good Standing   Ineligible
## 7            Critical   Ineligible
```

```
## 8       Good Standing   Ineligible
## 9       Good Standing   Ineligible
## 10      Good Standing   Ineligible
## 11           Critical   Ineligible
## 12           Critical   Ineligible
## 13      Good Standing   Ineligible
## 14      Good Standing   Ineligible
## 15            Warning   Ineligible
## 16      Good Standing   Ineligible
## 17      Good Standing   Ineligible
## 18            Warning   Ineligible
## 19            Warning   Ineligible
## 20      Good Standing   Ineligible
## 21           Critical   Ineligible
## 22      Good Standing     Eligible
## 23           Critical   Ineligible
## 24            Warning   Ineligible
## 25      Good Standing   Ineligible
## 26      Good Standing   Ineligible
## 27      Good Standing   Ineligible
## 28      Good Standing   Ineligible
## 29      Good Standing   Ineligible
## 30            Warning   Ineligible
## 31      Good Standing     Eligible
## 32      Good Standing   Ineligible
## 33      Good Standing   Ineligible
## 34            Warning   Ineligible
## 35            Warning   Ineligible
## 36      Good Standing   Ineligible
## 37      Good Standing   Ineligible
## 38            Warning   Ineligible
## 39      Good Standing   Ineligible
## 40            Warning   Ineligible
## 41            Warning     Eligible
## 42            Warning   Ineligible
## 43      Good Standing   Ineligible
## 44      Good Standing   Ineligible
## 45      Good Standing   Ineligible
## 46            Warning   Ineligible
## 47      Good Standing   Ineligible
## 48            Warning   Ineligible
## 49      Good Standing     Eligible
## 50            Warning   Ineligible
## 51            Warning     Eligible
## 52            Warning   Ineligible
## 53            Warning   Ineligible
## 54      Good Standing   Ineligible
## 55      Good Standing   Ineligible
## 56            Warning   Ineligible
## 57      Good Standing   Ineligible
## 58            Warning   Ineligible
## 59            Warning   Ineligible
## 60           Critical   Ineligible
## 61           Critical     Eligible
```

```
## 62      Good Standing   Ineligible
## 63            Warning   Ineligible
## 64      Good Standing   Ineligible
## 65            Warning   Ineligible
## 66      Good Standing   Ineligible
## 67            Warning   Ineligible
## 68           Critical   Ineligible
## 69      Good Standing   Ineligible
## 70            Warning   Ineligible
## 71            Warning   Ineligible
## 72            Warning   Ineligible
## 73      Good Standing   Ineligible
## 74            Warning     Eligible
## 75            Warning   Ineligible
## 76      Good Standing   Ineligible
## 77      Good Standing   Ineligible
## 78      Good Standing   Ineligible
## 79      Good Standing     Eligible
## 80            Warning   Ineligible
## 81      Good Standing   Ineligible
## 82           Critical   Ineligible
## 83      Good Standing   Ineligible
## 84            Warning   Ineligible
## 85      Good Standing   Ineligible
## 86      Good Standing   Ineligible
## 87            Warning   Ineligible
## 88            Warning   Ineligible
## 89      Good Standing   Ineligible
## 90            Warning   Ineligible
## 91           Critical   Ineligible
## 92           Critical   Ineligible
## 93      Good Standing   Ineligible
## 94            Warning   Ineligible
## 95      Good Standing   Ineligible
## 96           Critical   Ineligible
## 97      Good Standing   Ineligible
## 98            Warning   Ineligible
## 99      Good Standing   Ineligible
## 100     Good Standing   Ineligible
## 101          Critical   Ineligible
## 102     Good Standing   Ineligible
## 103     Good Standing   Ineligible
## 104          Critical   Ineligible
## 105          Critical   Ineligible
## 106     Good Standing   Ineligible
## 107     Good Standing   Ineligible
## 108     Good Standing   Ineligible
## 109           Warning   Ineligible
## 110          Critical   Ineligible
## 111     Good Standing   Ineligible
## 112     Good Standing   Ineligible
## 113          Critical   Ineligible
## 114           Warning   Ineligible
## 115           Warning   Ineligible
```

```
## 116          Critical   Ineligible
## 117           Warning   Ineligible
## 118     Good Standing   Ineligible
## 119          Critical   Ineligible
## 120     Good Standing   Ineligible
```

#6 .DEPARTMENTALHIGH ACHIEVERS COUNT

```r
#Filter for grade A ,group by department, and count
high_achievers_count<-df%>%
  filter(Grade=="A")%>%
  group_by(Department)%>%
  summarise(High_Achievers_Count=n())
print(high_achievers_count)
```

```
## # A tibble: 4 x 2
##   Department   High_Achievers_Count
##   <chr>                       <int>
## 1 Business                        3
## 2 Data Science                    6
## 3 Engineering                    10
## 4 IT                             11
```

#7. ACADEMIC RISK FLAG

```r
# %in% is used to check if a value exist within a specific vector or list of value
df<-df%>%
  mutate(Academic_Risk=case_when(
    Grade %in% c("D","F")~"At risk",
    Grade=="C"           ~"Stable",
    Grade %in% c("A","B")~"Excellent",
    TRUE                 ~"Other"
  ))
print(df)
```

```
##     Student_ID Gender   Department Age Assignment_Score Exam_Score
## 1            1   Male Data Science  27               80         57
## 2            2 Female           IT  24               74         61
## 3            3   Male     Business  29               58         36
## 4            4   Male  Engineering  26               87         51
## 5            5   Male           IT  24               55         67
## 6            6 Female Data Science  18               42         43
## 7            7   Male     Business  18               59         77
## 8            8   Male           IT  26               63         82
## 9            9   Male Data Science  28               93         73
## 10          10 Female Data Science  26               95         76
## 11          11   Male Data Science  21               72         60
## 12          12   Male           IT  26               63         84
## 13          13   Male           IT  20               91         59
## 14          14   Male           IT  24               50         58
## 15          15 Female  Engineering  23               88         47
## 16          16   Male           IT  25               47         94
```

```
## 17          17 Female           IT  28                75          41
## 18          18 Female           IT  26                77          91
## 19          19 Female  Engineering  22                79          70
## 20          20   Male           IT  18                59          79
## 21          21 Female Data Science  20                74          54
## 22          22   Male           IT  27                87          99
## 23          23 Female Data Science  29                64          42
## 24          24 Female Data Science  25                74          50
## 25          25 Female Data Science  28                64          48
## 26          26 Female  Engineering  23                68          49
## 27          27 Female  Engineering  25                57          66
## 28          28 Female  Engineering  26                85          97
## 29          29 Female           IT  21                57          85
## 30          30 Female Data Science  18                41          59
## 31          31   Male  Engineering  18                93          92
## 32          32   Male  Engineering  27                74          97
## 33          33 Female           IT  21                55          96
## 34          34 Female  Engineering  29                80          56
## 35          35 Female           IT  24                75          92
## 36          36   Male     Business  19                72          92
## 37          37 Female  Engineering  20                43          83
## 38          38   Male     Business  18                72          86
## 39          39   Male           IT  22                53          76
## 40          40   Male Data Science  18                60          49
## 41          41   Male  Engineering  25                87          88
## 42          42   Male           IT  18                59          94
## 43          43 Female Data Science  28                95          42
## 44          44 Female Data Science  18                47          87
## 45          45 Female     Business  19                46          94
## 46          46 Female           IT  19                42          39
## 47          47 Female Data Science  29                56          40
## 48          48   Male  Engineering  23                72          81
## 49          49 Female  Engineering  24                87          89
## 50          50 Female     Business  22                51          74
## 51          51   Male Data Science  18                98          86
## 52          52 Female           IT  18                90          50
## 53          53   Male  Engineering  20                61          47
## 54          54 Female Data Science  19                94          64
## 55          55   Male Data Science  22                61          53
## 56          56 Female     Business  29                85          51
## 57          57 Female  Engineering  27                69          97
## 58          58   Male  Engineering  23                77          53
## 59          59   Male           IT  24                77          92
## 60          60   Male  Engineering  21                84          89
## 61          61   Male           IT  24                90          96
## 62          62   Male  Engineering  28                93          57
## 63          63   Male     Business  25                47          43
## 64          64   Male  Engineering  28                66          46
## 65          65   Male           IT  18                66          35
## 66          66 Female           IT  23                73          92
## 67          67 Female     Business  25                60          35
## 68          68   Male  Engineering  22                69          68
## 69          69 Female  Engineering  21                72          82
## 70          70 Female     Business  19                67          35
```

```
## 71          71 Female  Engineering  23               86         50
## 72          72 Female  Engineering  23               72         95
## 73          73   Male           IT  28               44         98
## 74          74 Female  Engineering  18               99         97
## 75          75   Male  Engineering  26               87         56
## 76          76 Female     Business  29               58         60
## 77          77 Female     Business  28               43         50
## 78          78 Female Data Science  23               74         85
## 79          79   Male           IT  20               88         91
## 80          80 Female  Engineering  21               56         63
## 81          81   Male  Engineering  29               83         81
## 82          82 Female Data Science  28               67         96
## 83          83   Male  Engineering  21               69         50
## 84          84 Female           IT  20               68         82
## 85          85   Male Data Science  27               85         73
## 86          86   Male           IT  20               92         67
## 87          87 Female Data Science  20               45         57
## 88          88   Male Data Science  21               74         44
## 89          89 Female     Business  24               99         68
## 90          90 Female           IT  21               80         86
## 91          91 Female  Engineering  26               76         44
## 92          92 Female  Engineering  18               63         53
## 93          93 Female           IT  25               68         92
## 94          94 Female  Engineering  24               88         35
## 95          95 Female  Engineering  19               85         38
## 96          96 Female           IT  25               92         50
## 97          97 Female Data Science  18               70         58
## 98          98 Female           IT  28               74         36
## 99          99 Female Data Science  29               99         66
## 100        100   Male  Engineering  26               72         58
## 101        101   Male  Engineering  26               91         46
## 102        102 Female  Engineering  19               60         84
## 103        103 Female     Business  24               71         69
## 104        104 Female Data Science  27               62         67
## 105        105 Female     Business  20               72         67
## 106        106 Female     Business  24               42         95
## 107        107 Female           IT  29               57         85
## 108        108 Female     Business  27               64         77
## 109        109 Female           IT  26               81         46
## 110        110   Male           IT  21               70         99
## 111        111 Female     Business  29               93         67
## 112        112   Male Data Science  18               97         74
## 113        113 Female Data Science  19               42         77
## 114        114 Female Data Science  18               79         78
## 115        115   Male Data Science  29               85         63
## 116        116 Female Data Science  22               63         47
## 117        117   Male     Business  22               89         46
## 118        118 Female     Business  28               71         80
## 119        119 Female  Engineering  24               86         36
## 120        120   Male Data Science  26               61         69
##     Attendance_Percentage Total_Score Grade Honours_Category Retake_Required
## 1                      72         137     C      First Class              No
## 2                      66         135     C      First Class              No
## 3                      75          94     F      First Class              No
```

```
## 4                    57        138    C     First Class          No
## 5                    78        122    C     First Class          No
## 6                    75         85    F     First Class          No
## 7                    59        136    C     First Class          No
## 8                    75        145    B     First Class          No
## 9                    83        166    A     First Class          No
## 10                   90        171    A     First Class          No
## 11                   56        132    C     First Class          No
## 12                   53        147    B     First Class          No
## 13                   99        150    B     First Class          No
## 14                   94        108    F     First Class          No
## 15                   60        135    C     First Class          No
## 16                   78        141    B     First Class          No
## 17                   85        116    F     First Class          No
## 18                   74        168    A     First Class          No
## 19                   70        149    B     First Class          No
## 20                   85        138    B     First Class          No
## 21                   59        128    C     First Class          No
## 22                   86        186    A     First Class          No
## 23                   58        106    F     First Class          No
## 24                   73        124    F     First Class          No
## 25                   84        112    F     First Class          No
## 26                   98        117    F     First Class          No
## 27                   84        123    C     First Class          No
## 28                   97        182    A     First Class          No
## 29                   85        142    B     First Class          No
## 30                   67        100    F     First Class          No
## 31                   98        185    A     First Class          No
## 32                   88        171    A     First Class          No
## 33                   81        151    B     First Class          No
## 34                   73        136    C     First Class          No
## 35                   72        167    A     First Class          No
## 36                   81        164    A     First Class          No
## 37                   86        126    C     First Class          No
## 38                   61        158    A     First Class          No
## 39                   98        129    C     First Class          No
## 40                   62        109    F     First Class          No
## 41                   72        175    A     First Class          No
## 42                   74        153    A     First Class          No
## 43                   84        137    C     First Class          No
## 44                   90        134    B     First Class          No
## 45                   79        140    B     First Class          No
## 46                   66         81    F     First Class          No
## 47                   98         96    F     First Class          No
## 48                   69        153    B     First Class          No
## 49                   97        176    A     First Class          No
## 50                   74        125    C     First Class          No
## 51                   71        184    A     First Class          No
## 52                   62        140    C     First Class          No
## 53                   68        108    F     First Class          No
## 54                   98        158    B     First Class          No
## 55                   85        114    F     First Class          No
## 56                   61        136    C     First Class          No
## 57                   90        166    A     First Class          No
```

```
## 58                    68        130    C     First Class            No
## 59                    61        169    A     First Class            No
## 60                    58        173    A     First Class            No
## 61                    56        186    A     First Class            No
## 62                    77        150    B     First Class            No
## 63                    63         90    F     First Class            No
## 64                    80        112    F     First Class            No
## 65                    68        101    F     First Class            No
## 66                    96        165    A     First Class            No
## 67                    65         95    F     First Class            No
## 68                    54        137    C     First Class            No
## 69                    84        154    B     First Class            No
## 70                    61        102    F     First Class            No
## 71                    74        136    C     First Class            No
## 72                    70        167    A     First Class            No
## 73                    85        142    B     First Class            No
## 74                    72        196    A     First Class            No
## 75                    65        143    C     First Class            No
## 76                    88        118    F     First Class            No
## 77                    94         93    F     First Class            No
## 78                    91        159    A     First Class            No
## 79                    88        179    A     First Class            No
## 80                    63        119    C     First Class            No
## 81                    80        164    A     First Class            No
## 82                    54        163    A     First Class            No
## 83                    84        119    F     First Class            No
## 84                    72        150    B     First Class            No
## 85                    78        158    B     First Class            No
## 86                    92        159    B     First Class            No
## 87                    60        102    F     First Class            No
## 88                    67        118    F     First Class            No
## 89                    96        167    A     First Class            No
## 90                    61        166    A     First Class            No
## 91                    58        120    F     First Class            No
## 92                    59        116    F     First Class            No
## 93                    93        160    A     First Class            No
## 94                    66        123    F     First Class            No
## 95                    87        123    F     First Class            No
## 96                    56        142    C     First Class            No
## 97                    95        128    C     First Class            No
## 98                    62        110    F     First Class            No
## 99                    89        165    B     First Class            No
## 100                   91        130    C     First Class            No
## 101                   58        137    C     First Class            No
## 102                   99        144    B     First Class            No
## 103                   76        140    C     First Class            No
## 104                   51        129    C     First Class            No
## 105                   54        139    C     First Class            No
## 106                   78        137    B     First Class            No
## 107                   86        142    B     First Class            No
## 108                   87        141    B     First Class            No
## 109                   68        127    C     First Class            No
## 110                   57        169    A     First Class            No
## 111                   97        160    B     First Class            No
```

```
## 112                    94         171     A      First Class              No
## 113                    50         119     C      First Class              No
## 114                    71         157     B      First Class              No
## 115                    66         148     B      First Class              No
## 116                    56         110     F      First Class              No
## 117                    74         135     C      First Class              No
## 118                    94         151     B      First Class              No
## 119                    53         122     F      First Class              No
## 120                    85         130     C      First Class              No
##     Attendance_Status Extra_Credit Academic_Risk
## 1             Warning   Ineligible        Stable
## 2             Warning   Ineligible        Stable
## 3       Good Standing   Ineligible       At risk
## 4            Critical   Ineligible        Stable
## 5       Good Standing   Ineligible        Stable
## 6       Good Standing   Ineligible       At risk
## 7            Critical   Ineligible        Stable
## 8       Good Standing   Ineligible     Excellent
## 9       Good Standing   Ineligible     Excellent
## 10      Good Standing   Ineligible     Excellent
## 11           Critical   Ineligible        Stable
## 12           Critical   Ineligible     Excellent
## 13      Good Standing   Ineligible     Excellent
## 14      Good Standing   Ineligible       At risk
## 15            Warning   Ineligible        Stable
## 16      Good Standing   Ineligible     Excellent
## 17      Good Standing   Ineligible       At risk
## 18            Warning   Ineligible     Excellent
## 19            Warning   Ineligible     Excellent
## 20      Good Standing   Ineligible     Excellent
## 21           Critical   Ineligible        Stable
## 22      Good Standing     Eligible     Excellent
## 23           Critical   Ineligible       At risk
## 24            Warning   Ineligible       At risk
## 25      Good Standing   Ineligible       At risk
## 26      Good Standing   Ineligible       At risk
## 27      Good Standing   Ineligible        Stable
## 28      Good Standing   Ineligible     Excellent
## 29      Good Standing   Ineligible     Excellent
## 30            Warning   Ineligible       At risk
## 31      Good Standing     Eligible     Excellent
## 32      Good Standing   Ineligible     Excellent
## 33      Good Standing   Ineligible     Excellent
## 34            Warning   Ineligible        Stable
## 35            Warning   Ineligible     Excellent
## 36      Good Standing   Ineligible     Excellent
## 37      Good Standing   Ineligible        Stable
## 38            Warning   Ineligible     Excellent
## 39      Good Standing   Ineligible        Stable
## 40            Warning   Ineligible       At risk
## 41            Warning     Eligible     Excellent
## 42            Warning   Ineligible     Excellent
## 43      Good Standing   Ineligible        Stable
## 44      Good Standing   Ineligible     Excellent
```

```
## 45      Good Standing   Ineligible     Excellent
## 46            Warning   Ineligible       At risk
## 47      Good Standing   Ineligible       At risk
## 48            Warning   Ineligible     Excellent
## 49      Good Standing     Eligible     Excellent
## 50            Warning   Ineligible        Stable
## 51            Warning     Eligible     Excellent
## 52            Warning   Ineligible        Stable
## 53            Warning   Ineligible       At risk
## 54      Good Standing   Ineligible     Excellent
## 55      Good Standing   Ineligible       At risk
## 56            Warning   Ineligible        Stable
## 57      Good Standing   Ineligible     Excellent
## 58            Warning   Ineligible        Stable
## 59            Warning   Ineligible     Excellent
## 60           Critical   Ineligible     Excellent
## 61           Critical     Eligible     Excellent
## 62      Good Standing   Ineligible     Excellent
## 63            Warning   Ineligible       At risk
## 64      Good Standing   Ineligible       At risk
## 65            Warning   Ineligible       At risk
## 66      Good Standing   Ineligible     Excellent
## 67            Warning   Ineligible       At risk
## 68           Critical   Ineligible        Stable
## 69      Good Standing   Ineligible     Excellent
## 70            Warning   Ineligible       At risk
## 71            Warning   Ineligible        Stable
## 72            Warning   Ineligible     Excellent
## 73      Good Standing   Ineligible     Excellent
## 74            Warning     Eligible     Excellent
## 75            Warning   Ineligible        Stable
## 76      Good Standing   Ineligible       At risk
## 77      Good Standing   Ineligible       At risk
## 78      Good Standing   Ineligible     Excellent
## 79      Good Standing     Eligible     Excellent
## 80            Warning   Ineligible        Stable
## 81      Good Standing   Ineligible     Excellent
## 82           Critical   Ineligible     Excellent
## 83      Good Standing   Ineligible       At risk
## 84            Warning   Ineligible     Excellent
## 85      Good Standing   Ineligible     Excellent
## 86      Good Standing   Ineligible     Excellent
## 87            Warning   Ineligible       At risk
## 88            Warning   Ineligible       At risk
## 89      Good Standing   Ineligible     Excellent
## 90            Warning   Ineligible     Excellent
## 91           Critical   Ineligible       At risk
## 92           Critical   Ineligible       At risk
## 93      Good Standing   Ineligible     Excellent
## 94            Warning   Ineligible       At risk
## 95      Good Standing   Ineligible       At risk
## 96           Critical   Ineligible        Stable
## 97      Good Standing   Ineligible        Stable
## 98            Warning   Ineligible       At risk
```

```
## 99      Good Standing   Ineligible     Excellent
## 100     Good Standing   Ineligible        Stable
## 101          Critical   Ineligible        Stable
## 102     Good Standing   Ineligible     Excellent
## 103     Good Standing   Ineligible        Stable
## 104          Critical   Ineligible        Stable
## 105          Critical   Ineligible        Stable
## 106     Good Standing   Ineligible     Excellent
## 107     Good Standing   Ineligible     Excellent
## 108     Good Standing   Ineligible     Excellent
## 109           Warning   Ineligible        Stable
## 110          Critical   Ineligible     Excellent
## 111     Good Standing   Ineligible     Excellent
## 112     Good Standing   Ineligible     Excellent
## 113          Critical   Ineligible        Stable
## 114           Warning   Ineligible     Excellent
## 115           Warning   Ineligible     Excellent
## 116          Critical   Ineligible       At risk
## 117           Warning   Ineligible        Stable
## 118     Good Standing   Ineligible     Excellent
## 119          Critical   Ineligible       At risk
## 120     Good Standing   Ineligible        Stable
```

)) #8 STUDENT WHO IMPROVED IN EXAM

```r
# filter for rows where Exam Score is higher than Assignment Score
improved_student<-df%>%
  filter(Exam_Score>Assignment_Score)
print(improved_student)
```

```
##    Student_ID Gender   Department Age Assignment_Score Exam_Score
## 1           5   Male           IT  24               55         67
## 2           6 Female Data Science  18               42         43
## 3           7   Male     Business  18               59         77
## 4           8   Male           IT  26               63         82
## 5          12   Male           IT  26               63         84
## 6          14   Male           IT  24               50         58
## 7          16   Male           IT  25               47         94
## 8          18 Female           IT  26               77         91
## 9          20   Male           IT  18               59         79
## 10         22   Male           IT  27               87         99
## 11         27 Female  Engineering  25               57         66
## 12         28 Female  Engineering  26               85         97
## 13         29 Female           IT  21               57         85
## 14         30 Female Data Science  18               41         59
## 15         32   Male  Engineering  27               74         97
## 16         33 Female           IT  21               55         96
## 17         35 Female           IT  24               75         92
## 18         36   Male     Business  19               72         92
## 19         37 Female  Engineering  20               43         83
## 20         38   Male     Business  18               72         86
## 21         39   Male           IT  22               53         76
## 22         41   Male  Engineering  25               87         88
```

```
## 23         42   Male           IT  18               59         94
## 24         44 Female Data Science  18               47         87
## 25         45 Female     Business  19               46         94
## 26         48   Male  Engineering  23               72         81
## 27         49 Female  Engineering  24               87         89
## 28         50 Female     Business  22               51         74
## 29         57 Female  Engineering  27               69         97
## 30         59   Male           IT  24               77         92
## 31         60   Male  Engineering  21               84         89
## 32         61   Male           IT  24               90         96
## 33         66 Female           IT  23               73         92
## 34         69 Female  Engineering  21               72         82
## 35         72 Female  Engineering  23               72         95
## 36         73   Male           IT  28               44         98
## 37         76 Female     Business  29               58         60
## 38         77 Female     Business  28               43         50
## 39         78 Female Data Science  23               74         85
## 40         79   Male           IT  20               88         91
## 41         80 Female  Engineering  21               56         63
## 42         82 Female Data Science  28               67         96
## 43         84 Female           IT  20               68         82
## 44         87 Female Data Science  20               45         57
## 45         90 Female           IT  21               80         86
## 46         93 Female           IT  25               68         92
## 47        102 Female  Engineering  19               60         84
## 48        104 Female Data Science  27               62         67
## 49        106 Female     Business  24               42         95
## 50        107 Female           IT  29               57         85
## 51        108 Female     Business  27               64         77
## 52        110   Male           IT  21               70         99
## 53        113 Female Data Science  19               42         77
## 54        118 Female     Business  28               71         80
## 55        120   Male Data Science  26               61         69
##    Attendance_Percentage Total_Score Grade Honours_Category Retake_Required
## 1                     78         122     C      First Class              No
## 2                     75          85     F      First Class              No
## 3                     59         136     C      First Class              No
## 4                     75         145     B      First Class              No
## 5                     53         147     B      First Class              No
## 6                     94         108     F      First Class              No
## 7                     78         141     B      First Class              No
## 8                     74         168     A      First Class              No
## 9                     85         138     B      First Class              No
## 10                    86         186     A      First Class              No
## 11                    84         123     C      First Class              No
## 12                    97         182     A      First Class              No
## 13                    85         142     B      First Class              No
## 14                    67         100     F      First Class              No
## 15                    88         171     A      First Class              No
## 16                    81         151     B      First Class              No
## 17                    72         167     A      First Class              No
## 18                    81         164     A      First Class              No
## 19                    86         126     C      First Class              No
## 20                    61         158     A      First Class              No
```

```
## 21                    98         129     C      First Class              No
## 22                    72         175     A      First Class              No
## 23                    74         153     A      First Class              No
## 24                    90         134     B      First Class              No
## 25                    79         140     B      First Class              No
## 26                    69         153     B      First Class              No
## 27                    97         176     A      First Class              No
## 28                    74         125     C      First Class              No
## 29                    90         166     A      First Class              No
## 30                    61         169     A      First Class              No
## 31                    58         173     A      First Class              No
## 32                    56         186     A      First Class              No
## 33                    96         165     A      First Class              No
## 34                    84         154     B      First Class              No
## 35                    70         167     A      First Class              No
## 36                    85         142     B      First Class              No
## 37                    88         118     F      First Class              No
## 38                    94          93     F      First Class              No
## 39                    91         159     A      First Class              No
## 40                    88         179     A      First Class              No
## 41                    63         119     C      First Class              No
## 42                    54         163     A      First Class              No
## 43                    72         150     B      First Class              No
## 44                    60         102     F      First Class              No
## 45                    61         166     A      First Class              No
## 46                    93         160     A      First Class              No
## 47                    99         144     B      First Class              No
## 48                    51         129     C      First Class              No
## 49                    78         137     B      First Class              No
## 50                    86         142     B      First Class              No
## 51                    87         141     B      First Class              No
## 52                    57         169     A      First Class              No
## 53                    50         119     C      First Class              No
## 54                    94         151     B      First Class              No
## 55                    85         130     C      First Class              No
##    Attendance_Status Extra_Credit Academic_Risk
## 1      Good Standing   Ineligible        Stable
## 2      Good Standing   Ineligible       At risk
## 3           Critical   Ineligible        Stable
## 4      Good Standing   Ineligible     Excellent
## 5           Critical   Ineligible     Excellent
## 6      Good Standing   Ineligible       At risk
## 7      Good Standing   Ineligible     Excellent
## 8            Warning   Ineligible     Excellent
## 9      Good Standing   Ineligible     Excellent
## 10     Good Standing     Eligible     Excellent
## 11     Good Standing   Ineligible        Stable
## 12     Good Standing   Ineligible     Excellent
## 13     Good Standing   Ineligible     Excellent
## 14           Warning   Ineligible       At risk
## 15     Good Standing   Ineligible     Excellent
## 16     Good Standing   Ineligible     Excellent
## 17           Warning   Ineligible     Excellent
## 18     Good Standing   Ineligible     Excellent
```

```
## 19     Good Standing   Ineligible        Stable
## 20           Warning   Ineligible     Excellent
## 21     Good Standing   Ineligible        Stable
## 22           Warning     Eligible     Excellent
## 23           Warning   Ineligible     Excellent
## 24     Good Standing   Ineligible     Excellent
## 25     Good Standing   Ineligible     Excellent
## 26           Warning   Ineligible     Excellent
## 27     Good Standing     Eligible     Excellent
## 28           Warning   Ineligible        Stable
## 29     Good Standing   Ineligible     Excellent
## 30           Warning   Ineligible     Excellent
## 31          Critical   Ineligible     Excellent
## 32          Critical     Eligible     Excellent
## 33     Good Standing   Ineligible     Excellent
## 34     Good Standing   Ineligible     Excellent
## 35           Warning   Ineligible     Excellent
## 36     Good Standing   Ineligible     Excellent
## 37     Good Standing   Ineligible       At risk
## 38     Good Standing   Ineligible       At risk
## 39     Good Standing   Ineligible     Excellent
## 40     Good Standing     Eligible     Excellent
## 41           Warning   Ineligible        Stable
## 42          Critical   Ineligible     Excellent
## 43           Warning   Ineligible     Excellent
## 44           Warning   Ineligible       At risk
## 45           Warning   Ineligible     Excellent
## 46     Good Standing   Ineligible     Excellent
## 47     Good Standing   Ineligible     Excellent
## 48          Critical   Ineligible        Stable
## 49     Good Standing   Ineligible     Excellent
## 50     Good Standing   Ineligible     Excellent
## 51     Good Standing   Ineligible     Excellent
## 52          Critical   Ineligible     Excellent
## 53          Critical   Ineligible        Stable
## 54     Good Standing   Ineligible     Excellent
## 55     Good Standing   Ineligible        Stable
```

#9 CONDITIONAL BONUS MARKS

```r
# use if_else to add 5 marks only if attendance is 90 or above
conditional_bonus<-df%>%
  mutate(Adjusted_Total=if_else( Attendance_Percentage>=90,
                                 Total_Score+5,
                                 Total_Score))
print(conditional_bonus)
```

```
##     Student_ID Gender   Department Age Assignment_Score Exam_Score
## 1            1   Male Data Science  27               80         57
## 2            2 Female           IT  24               74         61
## 3            3   Male     Business  29               58         36
## 4            4   Male  Engineering  26               87         51
## 5            5   Male           IT  24               55         67
```

```
## 6            6 Female Data Science  18              42         43
## 7            7   Male     Business  18              59         77
## 8            8   Male           IT  26              63         82
## 9            9   Male Data Science  28              93         73
## 10          10 Female Data Science  26              95         76
## 11          11   Male Data Science  21              72         60
## 12          12   Male           IT  26              63         84
## 13          13   Male           IT  20              91         59
## 14          14   Male           IT  24              50         58
## 15          15 Female  Engineering  23              88         47
## 16          16   Male           IT  25              47         94
## 17          17 Female           IT  28              75         41
## 18          18 Female           IT  26              77         91
## 19          19 Female  Engineering  22              79         70
## 20          20   Male           IT  18              59         79
## 21          21 Female Data Science  20              74         54
## 22          22   Male           IT  27              87         99
## 23          23 Female Data Science  29              64         42
## 24          24 Female Data Science  25              74         50
## 25          25 Female Data Science  28              64         48
## 26          26 Female  Engineering  23              68         49
## 27          27 Female  Engineering  25              57         66
## 28          28 Female  Engineering  26              85         97
## 29          29 Female           IT  21              57         85
## 30          30 Female Data Science  18              41         59
## 31          31   Male  Engineering  18              93         92
## 32          32   Male  Engineering  27              74         97
## 33          33 Female           IT  21              55         96
## 34          34 Female  Engineering  29              80         56
## 35          35 Female           IT  24              75         92
## 36          36   Male     Business  19              72         92
## 37          37 Female  Engineering  20              43         83
## 38          38   Male     Business  18              72         86
## 39          39   Male           IT  22              53         76
## 40          40   Male Data Science  18              60         49
## 41          41   Male  Engineering  25              87         88
## 42          42   Male           IT  18              59         94
## 43          43 Female Data Science  28              95         42
## 44          44 Female Data Science  18              47         87
## 45          45 Female     Business  19              46         94
## 46          46 Female           IT  19              42         39
## 47          47 Female Data Science  29              56         40
## 48          48   Male  Engineering  23              72         81
## 49          49 Female  Engineering  24              87         89
## 50          50 Female     Business  22              51         74
## 51          51   Male Data Science  18              98         86
## 52          52 Female           IT  18              90         50
## 53          53   Male  Engineering  20              61         47
## 54          54 Female Data Science  19              94         64
## 55          55   Male Data Science  22              61         53
## 56          56 Female     Business  29              85         51
## 57          57 Female  Engineering  27              69         97
## 58          58   Male  Engineering  23              77         53
## 59          59   Male           IT  24              77         92
```

```
## 60          60   Male  Engineering  21                84          89
## 61          61   Male           IT  24                90          96
## 62          62   Male  Engineering  28                93          57
## 63          63   Male     Business  25                47          43
## 64          64   Male  Engineering  28                66          46
## 65          65   Male           IT  18                66          35
## 66          66 Female           IT  23                73          92
## 67          67 Female     Business  25                60          35
## 68          68   Male  Engineering  22                69          68
## 69          69 Female  Engineering  21                72          82
## 70          70 Female     Business  19                67          35
## 71          71 Female  Engineering  23                86          50
## 72          72 Female  Engineering  23                72          95
## 73          73   Male           IT  28                44          98
## 74          74 Female  Engineering  18                99          97
## 75          75   Male  Engineering  26                87          56
## 76          76 Female     Business  29                58          60
## 77          77 Female     Business  28                43          50
## 78          78 Female Data Science  23                74          85
## 79          79   Male           IT  20                88          91
## 80          80 Female  Engineering  21                56          63
## 81          81   Male  Engineering  29                83          81
## 82          82 Female Data Science  28                67          96
## 83          83   Male  Engineering  21                69          50
## 84          84 Female           IT  20                68          82
## 85          85   Male Data Science  27                85          73
## 86          86   Male           IT  20                92          67
## 87          87 Female Data Science  20                45          57
## 88          88   Male Data Science  21                74          44
## 89          89 Female     Business  24                99          68
## 90          90 Female           IT  21                80          86
## 91          91 Female  Engineering  26                76          44
## 92          92 Female  Engineering  18                63          53
## 93          93 Female           IT  25                68          92
## 94          94 Female  Engineering  24                88          35
## 95          95 Female  Engineering  19                85          38
## 96          96 Female           IT  25                92          50
## 97          97 Female Data Science  18                70          58
## 98          98 Female           IT  28                74          36
## 99          99 Female Data Science  29                99          66
## 100        100   Male  Engineering  26                72          58
## 101        101   Male  Engineering  26                91          46
## 102        102 Female  Engineering  19                60          84
## 103        103 Female     Business  24                71          69
## 104        104 Female Data Science  27                62          67
## 105        105 Female     Business  20                72          67
## 106        106 Female     Business  24                42          95
## 107        107 Female           IT  29                57          85
## 108        108 Female     Business  27                64          77
## 109        109 Female           IT  26                81          46
## 110        110   Male           IT  21                70          99
## 111        111 Female     Business  29                93          67
## 112        112   Male Data Science  18                97          74
## 113        113 Female Data Science  19                42          77
```

```
## 114        114 Female Data Science  18               79         78
## 115        115   Male Data Science  29               85         63
## 116        116 Female Data Science  22               63         47
## 117        117   Male     Business  22               89         46
## 118        118 Female     Business  28               71         80
## 119        119 Female  Engineering  24               86         36
## 120        120   Male Data Science  26               61         69
##     Attendance_Percentage Total_Score Grade Honours_Category Retake_Required
## 1                      72         137     C      First Class              No
## 2                      66         135     C      First Class              No
## 3                      75          94     F      First Class              No
## 4                      57         138     C      First Class              No
## 5                      78         122     C      First Class              No
## 6                      75          85     F      First Class              No
## 7                      59         136     C      First Class              No
## 8                      75         145     B      First Class              No
## 9                      83         166     A      First Class              No
## 10                     90         171     A      First Class              No
## 11                     56         132     C      First Class              No
## 12                     53         147     B      First Class              No
## 13                     99         150     B      First Class              No
## 14                     94         108     F      First Class              No
## 15                     60         135     C      First Class              No
## 16                     78         141     B      First Class              No
## 17                     85         116     F      First Class              No
## 18                     74         168     A      First Class              No
## 19                     70         149     B      First Class              No
## 20                     85         138     B      First Class              No
## 21                     59         128     C      First Class              No
## 22                     86         186     A      First Class              No
## 23                     58         106     F      First Class              No
## 24                     73         124     F      First Class              No
## 25                     84         112     F      First Class              No
## 26                     98         117     F      First Class              No
## 27                     84         123     C      First Class              No
## 28                     97         182     A      First Class              No
## 29                     85         142     B      First Class              No
## 30                     67         100     F      First Class              No
## 31                     98         185     A      First Class              No
## 32                     88         171     A      First Class              No
## 33                     81         151     B      First Class              No
## 34                     73         136     C      First Class              No
## 35                     72         167     A      First Class              No
## 36                     81         164     A      First Class              No
## 37                     86         126     C      First Class              No
## 38                     61         158     A      First Class              No
## 39                     98         129     C      First Class              No
## 40                     62         109     F      First Class              No
## 41                     72         175     A      First Class              No
## 42                     74         153     A      First Class              No
## 43                     84         137     C      First Class              No
## 44                     90         134     B      First Class              No
## 45                     79         140     B      First Class              No
## 46                     66          81     F      First Class              No
```

```
## 47                    98          96     F      First Class              No
## 48                    69         153     B      First Class              No
## 49                    97         176     A      First Class              No
## 50                    74         125     C      First Class              No
## 51                    71         184     A      First Class              No
## 52                    62         140     C      First Class              No
## 53                    68         108     F      First Class              No
## 54                    98         158     B      First Class              No
## 55                    85         114     F      First Class              No
## 56                    61         136     C      First Class              No
## 57                    90         166     A      First Class              No
## 58                    68         130     C      First Class              No
## 59                    61         169     A      First Class              No
## 60                    58         173     A      First Class              No
## 61                    56         186     A      First Class              No
## 62                    77         150     B      First Class              No
## 63                    63          90     F      First Class              No
## 64                    80         112     F      First Class              No
## 65                    68         101     F      First Class              No
## 66                    96         165     A      First Class              No
## 67                    65          95     F      First Class              No
## 68                    54         137     C      First Class              No
## 69                    84         154     B      First Class              No
## 70                    61         102     F      First Class              No
## 71                    74         136     C      First Class              No
## 72                    70         167     A      First Class              No
## 73                    85         142     B      First Class              No
## 74                    72         196     A      First Class              No
## 75                    65         143     C      First Class              No
## 76                    88         118     F      First Class              No
## 77                    94          93     F      First Class              No
## 78                    91         159     A      First Class              No
## 79                    88         179     A      First Class              No
## 80                    63         119     C      First Class              No
## 81                    80         164     A      First Class              No
## 82                    54         163     A      First Class              No
## 83                    84         119     F      First Class              No
## 84                    72         150     B      First Class              No
## 85                    78         158     B      First Class              No
## 86                    92         159     B      First Class              No
## 87                    60         102     F      First Class              No
## 88                    67         118     F      First Class              No
## 89                    96         167     A      First Class              No
## 90                    61         166     A      First Class              No
## 91                    58         120     F      First Class              No
## 92                    59         116     F      First Class              No
## 93                    93         160     A      First Class              No
## 94                    66         123     F      First Class              No
## 95                    87         123     F      First Class              No
## 96                    56         142     C      First Class              No
## 97                    95         128     C      First Class              No
## 98                    62         110     F      First Class              No
## 99                    89         165     B      First Class              No
## 100                   91         130     C      First Class              No
```

```
## 101                    58         137     C      First Class              No
## 102                    99         144     B      First Class              No
## 103                    76         140     C      First Class              No
## 104                    51         129     C      First Class              No
## 105                    54         139     C      First Class              No
## 106                    78         137     B      First Class              No
## 107                    86         142     B      First Class              No
## 108                    87         141     B      First Class              No
## 109                    68         127     C      First Class              No
## 110                    57         169     A      First Class              No
## 111                    97         160     B      First Class              No
## 112                    94         171     A      First Class              No
## 113                    50         119     C      First Class              No
## 114                    71         157     B      First Class              No
## 115                    66         148     B      First Class              No
## 116                    56         110     F      First Class              No
## 117                    74         135     C      First Class              No
## 118                    94         151     B      First Class              No
## 119                    53         122     F      First Class              No
## 120                    85         130     C      First Class              No
##     Attendance_Status Extra_Credit Academic_Risk Adjusted_Total
## 1             Warning   Ineligible        Stable            137
## 2             Warning   Ineligible        Stable            135
## 3       Good Standing   Ineligible       At risk             94
## 4            Critical   Ineligible        Stable            138
## 5       Good Standing   Ineligible        Stable            122
## 6       Good Standing   Ineligible       At risk             85
## 7            Critical   Ineligible        Stable            136
## 8       Good Standing   Ineligible     Excellent            145
## 9       Good Standing   Ineligible     Excellent            166
## 10      Good Standing   Ineligible     Excellent            176
## 11           Critical   Ineligible        Stable            132
## 12           Critical   Ineligible     Excellent            147
## 13      Good Standing   Ineligible     Excellent            155
## 14      Good Standing   Ineligible       At risk            113
## 15            Warning   Ineligible        Stable            135
## 16      Good Standing   Ineligible     Excellent            141
## 17      Good Standing   Ineligible       At risk            116
## 18            Warning   Ineligible     Excellent            168
## 19            Warning   Ineligible     Excellent            149
## 20      Good Standing   Ineligible     Excellent            138
## 21           Critical   Ineligible        Stable            128
## 22      Good Standing     Eligible     Excellent            186
## 23           Critical   Ineligible       At risk            106
## 24            Warning   Ineligible       At risk            124
## 25      Good Standing   Ineligible       At risk            112
## 26      Good Standing   Ineligible       At risk            122
## 27      Good Standing   Ineligible        Stable            123
## 28      Good Standing   Ineligible     Excellent            187
## 29      Good Standing   Ineligible     Excellent            142
## 30            Warning   Ineligible       At risk            100
## 31      Good Standing     Eligible     Excellent            190
## 32      Good Standing   Ineligible     Excellent            171
## 33      Good Standing   Ineligible     Excellent            151
```

```
## 34            Warning   Ineligible        Stable            136
## 35            Warning   Ineligible     Excellent            167
## 36      Good Standing   Ineligible     Excellent            164
## 37      Good Standing   Ineligible        Stable            126
## 38            Warning   Ineligible     Excellent            158
## 39      Good Standing   Ineligible        Stable            134
## 40            Warning   Ineligible       At risk            109
## 41            Warning     Eligible     Excellent            175
## 42            Warning   Ineligible     Excellent            153
## 43      Good Standing   Ineligible        Stable            137
## 44      Good Standing   Ineligible     Excellent            139
## 45      Good Standing   Ineligible     Excellent            140
## 46            Warning   Ineligible       At risk             81
## 47      Good Standing   Ineligible       At risk            101
## 48            Warning   Ineligible     Excellent            153
## 49      Good Standing     Eligible     Excellent            181
## 50            Warning   Ineligible        Stable            125
## 51            Warning     Eligible     Excellent            184
## 52            Warning   Ineligible        Stable            140
## 53            Warning   Ineligible       At risk            108
## 54      Good Standing   Ineligible     Excellent            163
## 55      Good Standing   Ineligible       At risk            114
## 56            Warning   Ineligible        Stable            136
## 57      Good Standing   Ineligible     Excellent            171
## 58            Warning   Ineligible        Stable            130
## 59            Warning   Ineligible     Excellent            169
## 60           Critical   Ineligible     Excellent            173
## 61           Critical     Eligible     Excellent            186
## 62      Good Standing   Ineligible     Excellent            150
## 63            Warning   Ineligible       At risk             90
## 64      Good Standing   Ineligible       At risk            112
## 65            Warning   Ineligible       At risk            101
## 66      Good Standing   Ineligible     Excellent            170
## 67            Warning   Ineligible       At risk             95
## 68           Critical   Ineligible        Stable            137
## 69      Good Standing   Ineligible     Excellent            154
## 70            Warning   Ineligible       At risk            102
## 71            Warning   Ineligible        Stable            136
## 72            Warning   Ineligible     Excellent            167
## 73      Good Standing   Ineligible     Excellent            142
## 74            Warning     Eligible     Excellent            196
## 75            Warning   Ineligible        Stable            143
## 76      Good Standing   Ineligible       At risk            118
## 77      Good Standing   Ineligible       At risk             98
## 78      Good Standing   Ineligible     Excellent            164
## 79      Good Standing     Eligible     Excellent            179
## 80            Warning   Ineligible        Stable            119
## 81      Good Standing   Ineligible     Excellent            164
## 82           Critical   Ineligible     Excellent            163
## 83      Good Standing   Ineligible       At risk            119
## 84            Warning   Ineligible     Excellent            150
## 85      Good Standing   Ineligible     Excellent            158
## 86      Good Standing   Ineligible     Excellent            164
## 87            Warning   Ineligible       At risk            102
```

```
## 88            Warning   Ineligible       At risk            118
## 89      Good Standing   Ineligible     Excellent            172
## 90            Warning   Ineligible     Excellent            166
## 91           Critical   Ineligible       At risk            120
## 92           Critical   Ineligible       At risk            116
## 93      Good Standing   Ineligible     Excellent            165
## 94            Warning   Ineligible       At risk            123
## 95      Good Standing   Ineligible       At risk            123
## 96           Critical   Ineligible        Stable            142
## 97      Good Standing   Ineligible        Stable            133
## 98            Warning   Ineligible       At risk            110
## 99      Good Standing   Ineligible     Excellent            165
## 100     Good Standing   Ineligible        Stable            135
## 101          Critical   Ineligible        Stable            137
## 102     Good Standing   Ineligible     Excellent            149
## 103     Good Standing   Ineligible        Stable            140
## 104          Critical   Ineligible        Stable            129
## 105          Critical   Ineligible        Stable            139
## 106     Good Standing   Ineligible     Excellent            137
## 107     Good Standing   Ineligible     Excellent            142
## 108     Good Standing   Ineligible     Excellent            141
## 109           Warning   Ineligible        Stable            127
## 110          Critical   Ineligible     Excellent            169
## 111     Good Standing   Ineligible     Excellent            165
## 112     Good Standing   Ineligible     Excellent            176
## 113          Critical   Ineligible        Stable            119
## 114           Warning   Ineligible     Excellent            157
## 115           Warning   Ineligible     Excellent            148
## 116          Critical   Ineligible       At risk            110
## 117           Warning   Ineligible        Stable            135
## 118     Good Standing   Ineligible     Excellent            156
## 119          Critical   Ineligible       At risk            122
## 120     Good Standing   Ineligible        Stable            130
```

10 OVERALL PERFOMANCE SUMMARY BY GENDER

```r
# group by gender and calculate pass/fail total
perfomance_summary<-df%>%
  group_by(Gender)%>%
  summarise(
    Number_Passed=sum(Total_Score>=50,na.rm = TRUE),
    Number_Failed=sum(Total_Score<50,na.rm = TRUE)
  )
print(perfomance_summary)
```

```
## # A tibble: 2 x 3
##   Gender Number_Passed Number_Failed
##   <chr>          <int>         <int>
## 1 Female            70             0
## 2 Male              50             0
```